



École Polytechnique Fédérale de Lausanne

The Development of Property Induction
During Language Model Pre-training

by Jan Sobotka

Master Thesis

Approved by the Examining Committee:

Prof. Dr. Antoine Bosselut
Thesis Advisor

Dr. Xavier Renard
External Expert

Dr. Xavier Renard
Thesis Supervisor

AXA Group Operations
81 Rue Mstislav Rostropovitch
75017 Paris
France

August 14, 2026

Acknowledgments

First and foremost, I would like to thank my family for their support throughout my studies, which allowed me to dedicate my full attention to my education and hobbies. They have been with me through both ups and downs, and I am profoundly grateful for it.

I would also like to thank the Bakala Foundation for providing two years of financial support, without which my studies at EPFL in Switzerland would not have been possible.

I am also very grateful to my supervisors, Prof. Dr. Antoine Bosselut and Dr. Xavier Renard, for their guidance and expertise throughout this thesis project. I especially appreciated my discussions with Dr. Renard about career choices, experiences in industry versus academia, and science more broadly. Finally, a big thank you to Ekaterina Goliakova for her ideas, advice, and good coffee conversations.

This thesis was typeset using the $\text{\LaTeX}$ template provided by the HexHive laboratory at EPFL (https://github.com/HexHive/thesis_template).

I acknowledge the use of AI (Claude Opus 5; Anthropic) during the preparation of this thesis, under full human supervision. Its use was limited to providing suggestions for coding, proofreading, and refining the academic language of my own work. All content was reviewed, verified, and edited by the author.

Lausanne, August 14, 2026

Jan Sobotka

Abstract

People generalize a newly learned property from one entity to others in proportion to how taxonomically similar they are. For example, told that robins have sesamoid bones, people extend the property to sparrows much more easily than to whales. This behavior, called category-based property induction, has been studied in people for decades. Whether it also develops in language models (LMs) trained only to predict the next token, and how it changes over the course of training, remains poorly understood. Existing evaluations of LMs typically examine a single trained checkpoint, lack control over the training data, or rely on training objectives that differ from standard pre-training. We address these gaps by constructing a synthetic taxonomy of novel entities and properties, and injecting it into the continued pre-training of Pythia and OLMo 2 models ranging from 70M to 7B parameters. With this controlled yet realistic setup, we minimize the effects of unaccounted confounders and ensure that the measured mechanism stays close to the one that operates in practice.

Our experiments show that language models develop taxonomically aligned inductive generalization from next-token prediction alone. A property learned for one entity transfers more strongly to entities in the same category than to entities in other categories. By inspecting the model’s internals, we find that entities that share a deeper category become more similar in the model’s activation space, and the full taxonomic tree can be recovered from entity representations by clustering. We further find that induction strengthens with the amount of pre-training that precedes the injected synthetic data, while increasing model size at a fixed data budget weakens it. Finally, we show that a taxonomy stored in the weights lets models generalize new properties supplied in context, and that when the context implies a conflicting taxonomy, the context wins. Together, these results show that the inductive behavior of LMs is a result of a latent structure in the data, with implications for data curation and for predicting how far learned facts will spread.

Contents

Acknowledgments	2
Abstract	3
1 Introduction	6
2 Background	10
2.1 Category-based property induction	10
2.1.1 The inductive argument and its strength	10
2.1.2 Phenomena of human induction	11
2.2 Language models and next-token prediction	11
2.3 The Transformer architecture	12
2.4 Representational geometry	13
2.4.1 Concepts as directions	13
2.4.2 Similarity between concept representations	13
2.5 Memorization, generalization, and scale	14
2.5.1 Two ways of fitting the same data	14
2.5.2 The effect of scale	14
2.5.3 In-context learning	15
2.6 From token probabilities to property induction	15
3 Experimental Setup	17
3.1 Models	17
3.1.1 Pythia	18
3.1.2 OLMo 2	19
3.2 Training Data	19
3.2.1 Synthetic taxonomy of concepts	19
3.2.2 Data mixing	22
3.3 Evaluation	23
3.3.1 Constraining the completion space	23

4 Experiments	25
4.1 LMs inductively generalize along the latent taxonomy	26
4.2 The taxonomy is recoverable from entity representations	28
4.3 The effect of batch size	30
4.4 Development and the effect of scale	31
4.5 Flexibility and interference across taxonomies	34
4.6 In-weights taxonomy shapes in-context induction	35
5 Related Work	38
5.1 Category-based induction in cognitive science	38
5.2 Property induction in language models	39
5.3 Learning dynamics and semantic development	39
5.4 Controlled injection and the reach of a learned fact	40
5.5 The geometry of concepts	41
6 Conclusion	42
6.1 Summary of findings	42
6.2 Limitations	44
6.3 Future directions	44
6.4 Closing remarks	45
Bibliography	47

Chapter 1

Introduction

Human semantic cognition is organized around categories composed of related entities [55]. We do not store the properties of every object independently; instead, we infer them from category membership. This lets us predict unobserved features of novel or partially observed entities [55]. Such an inference is called *category-based property induction*: learning that a novel property holds for one entity, we extend it more strongly to taxonomically close entities than to distant ones. For example, when told that robins have sesamoid bones, we expect sparrows to have them too, but are far less sure about whales [46].

Importantly, this type of induction is sensitive to *which* taxonomy is being used. The same two objects may be close under one organizing scheme and distant under another. For example, a whale and a shark are similar as swimming things and dissimilar as biological creatures, one a mammal and the other a fish. The categorical structure considered by the reasoner decides the measurement of similarity, as well as the direction of inference.

Induction is therefore flexible rather than fixed: it depends on a particular learned organization of the world. This flexibility is reflected in the fact that the taxonomy supporting induction is itself acquired: over development, children shift from generalizing based on perceptual similarity to generalizing based on category membership [3, 16]. More broadly, property knowledge differentiates into a taxonomic hierarchy over years of exposure, with the resulting structure supporting increasingly systematic induction throughout child development [26, 34, 35]. This systematicity is reflected in the finding that broad distinctions between entities are acquired before fine ones, so a child separates animals from artifacts before separating robins from sparrows [35, 49].

Saxe, McClelland, and Ganguli [57] show that the same coarse-to-fine ordering emerges in deep linear networks trained with gradient descent. They derive closed-form learning trajectories in which the modes of the input–output correlation matrix are learned in order of decreasing singular value, so the coarsest distinctions in the data are captured first and finer ones follow. The theory predicts

progressive differentiation, stage-like transitions in the learning curve, and a representational geometry in which the similarity between two items reflects the depth of their shared category.

However, the theory of [57] is about idealized linear networks trained on idealized synthetic data, and it has not been tested in the setting that now dominates machine learning: large language models (LLMs) based on the Transformer architecture [62]. LLMs, which compose many nonlinear transformations, are traditionally pre-trained by predicting the next token in natural text (Chapter 2), so it is unclear how well results from [57] transfer to this setting. Whether taxonomic structure and category-based induction emerge in LMs¹ is therefore the main topic of this thesis. Studying it is important for two reasons.

Shared categories underlie communication. An LLM assistant is only useful if its sense of which things belong together matches its user’s. When a user asks for something to be applied “to the rest of them”, or asks whether a finding about one item transfers to another, the answer depends on a categorical judgment by the model. Mismatched taxonomies may produce failures that are hard to diagnose, because the model’s answer is fluent and its organizing assumption is never stated.

Induction determines what a fact does to a model. From a practical as well as scientific standpoint, we would like to predict how a model will generalize after training on a given corpus. Adding a document does not only add the facts it states; it moves predictions about every entity that the model treats as similar to the ones mentioned. Induction is the mechanism of that (over)generalization, so understanding it is a prerequisite for reasoning about data curation, for anticipating model behavior, and for knowing whether a targeted training edit will stay local or propagate. Property induction allows for clean testing of this type of generalization.

While the widespread use of LLMs and therefore the importance of fully understanding them has been increasing [2, 6, 12], property induction, as originally conceptualized in cognitive science [26, 34, 35, 49, 53], has been mostly studied in people. Only recently have researchers begun to analyze property induction in LMs [21, 38, 54]. Three gaps in this line of work motivate our study.

No developmental measurement. Most existing evaluations take a single checkpoint and ask whether induction is present at that point [21, 54]. That methodology completely leaves out the dynamics: when during training does induction appear, how does it strengthen, how does it interact with memorization, how does it depend on model scale and on how much pre-training preceded the evidence. A static evaluation cannot answer any of these, and they are exactly the questions the developmental literature, such as the theory of Saxe, McClelland, and Ganguli [57], can bring.

¹We reserve LLM for frontier-scale systems.

Leakage. Studies that evaluate pre-trained models using real entities and plausible-sounding properties cannot exclude the possibility that the model encountered the target fact, or a paraphrase of it, somewhere in its pre-training corpus [21, 38, 54]. When a model asserts that sparrows have sesamoid bones, we cannot tell whether it generalized from robins or was directly trained on it. Because pre-training corpora are large and only partially documented, this confound is difficult to rule out, and it is core to the specific claim being made: induction and recall are exactly what needs to be distinguished. Additionally, such studies have little control and limited knowledge about the ground-truth taxonomy or the frequency of exposure to various entity-property pairs.

Objective mismatch. Work that trains models explicitly for property induction fine-tunes them to answer yes/no questions [38]. This measures whether a model *can* be taught to report property judgments, but not whether such structure arises from the objective under which most currently deployed LLMs actually acquire most of their semantic knowledge. Since our question is what standard pre-training gives rise to on its own, the training signal has to remain next-token prediction.

We close these gaps with a partially synthetic setup that remains realistic in its training regime. We construct a controlled taxonomy of novel concepts, render it into natural-language text, and inject it into the pre-training of real LMs ranging from 70M to 7B parameters alongside their original pre-training corpora. Every entity and property is a pronounceable pseudoword, so entities carry little to no prior association and every co-occurrence in the model’s experience is one we placed there ourselves. Leakage is highly unlikely by construction. At the same time, the model continues to be trained by ordinary next-token prediction on natural text, so we are measuring the mechanism that operates in practice rather than a highly idealized proxy for it. Additionally, because we control the taxonomy, we know the ground truth against which generalization should be judged, and because we evaluate throughout training, we observe development rather than a single endpoint.

This setup allows us to ask four questions:

- RQ1** Do language models develop taxonomically aligned inductive generalization from next-token prediction alone, and does it depend on the taxonomic structure of the data?
- RQ2** If so, what internal changes in the model support it? Does the latent taxonomy become recoverable from the model’s representations, as Saxe, McClelland, and Ganguli [57] predict?
- RQ3** How does inductive generalization depend on training: on the amount of continued pre-training, on how much pre-training preceded the injected evidence, and on model scale?
- RQ4** How does induction stored in weights interact with evidence supplied in context, particularly when the two disagree?

By answering these questions, we make the following contributions:

1. We show that LMs develop taxonomically aligned inductive generalization purely from next-token prediction, supported by a latent taxonomy in the training data. Removing the taxonomic signal removes the generalization (Section 4.1).
2. This generalization is reflected in the model’s internal representations. Entity representations become more similar the deeper their shared category, and the full taxonomy is recoverable from them by clustering, empirically validating the theory of Saxe, McClelland, and Ganguli [57] (Section 4.2).
3. We find that the strength of acquired induction grows with continued pre-training and with the pre-training stage of the base checkpoint. Counterintuitively, given the same data budget, increasing model size *reduces* correct induction (Section 4.4).
4. Our experiments show that learning only the shared properties of a taxonomy is enough to enable in-context inductive generalization over novel properties, and when an in-context taxonomy conflicts with the one in weights, the model generalizes according to the context (Section 4.6).

Two limitations of the thesis are worth stating at the beginning. First, the taxonomy we inject is small, balanced, and noiseless, whereas real conceptual structure is large and difficult to write down exactly. This allows for control that makes the measurement interpretable, at the cost of realism. Second, the models we study span 70M to 7B parameters, which allows for a scaling comparison but does not establish that the trends continue at the scale of currently deployed systems. We return to both points in Chapter 6.

The structure of the thesis is as follows. Chapter 2 introduces the necessary background on property induction, language models, the Transformer architecture, next-token prediction, and in-context learning. Chapter 3 describes the models, the synthetic taxonomy, the data mixing and injection procedure, and the evaluation metrics, including the induction advantage used throughout. Chapter 4 presents the results, addressing RQ1–RQ4: the emergence of taxonomic induction and its ablation, the representation analysis and tree recovery, the effects of training duration and model scale, the interference between conflicting taxonomies, and the interaction between in-weights and in-context induction. Chapter 5 further contextualizes the work in the literature, and Chapter 6 summarizes the main findings and concludes with potential future directions.

Chapter 2

Background

This section introduces the minimum needed to follow the rest of the thesis: a formalization of property induction, what a language model is, the architecture that dominates modern systems, the objective these models are trained on, and ways in which they can incorporate and use new information.

2.1 Category-based property induction

The behavior this thesis measures was defined and characterized in cognitive science, with definitions that fix what counts as *correct* generalization. This section states the phenomenon and describes the characteristic empirical regularities.

2.1.1 The inductive argument and its strength

Category-based property induction is studied through arguments of the form

Robins have sesamoid bones.	_____
Sparrows have sesamoid bones.	

where the statements above the line are *premises* and the statement below is the *conclusion*. Each statement links a property to a category or an individual. We use the terms individual, subject, and entity interchangeably. The studied quantity is the *argument strength*: the degree to which a reasoner who accepts the premises is willing to accept the conclusion [46, 53]. Strength is graded rather than binary, which is what makes it comparable between people or language models.

Two features of the standard design are worth emphasizing because our setup imitates both. First, the projected property is a *blank predicate*: a property such as “has sesamoid bones” that individuals have no knowledge about, chosen such that judgments cannot come from what the reasoner already knows about the property and must instead come from the relation between the categories or entities [46]. Second, the categories and the underlying taxonomy are familiar to the individuals. In our synthetic setup, properties and entities are pseudowords, so the property is blank *and* the taxonomy is one we design and present to the LMs ourselves. This is what makes the ground truth known, leakage highly unlikely, and ensures that the models are familiar with it.

2.1.2 Phenomena of human induction

Two of the most robust regularities across human studies are the *similarity* and the *conclusion specificity* phenomena [22, 41, 46]:

- **Similarity.** Argument strength increases with the similarity between premise and conclusion categories. *Robin* $\rightarrow$ *sparrow* is stronger than *robin* $\rightarrow$ *whale*.
- **Conclusion specificity.** For fixed premises, a more specific conclusion category is easier to accept than a broader one. For example, projecting from robins to sparrows (*i.e.*, other birds) is more common than projecting from robins to all animals.

Importantly, these phenomena are stated in terms of a taxonomy the reasoner knows; nothing about them is intrinsic only to the entities. Our experimental design targets both. We evaluate the similarity phenomenon by comparing the within-category versus across-category generalization (Section 3.3) with a known taxonomic tree, and we consider the graded effect of the deepest shared level in the tree as the conclusion specificity.

2.2 Language models and next-token prediction

A language model is a probability distribution over sequences of text. The text is first split into a sequence of *tokens* $x = (x_1, \dots, x_T)$, each drawn from a fixed vocabulary $\mathcal{V}$. Tokens are usually sub-word units produced by an algorithm such as byte-pair encoding [58], so that common words are single tokens and rare words are split into pieces. A model with parameters θ assigns a probability to the whole sequence through the chain rule,

$$p_{\theta}(x) = \prod_{t=1}^T p_{\theta}(x_t | x_{<t}), \quad (2.1)$$

where $x_{<t} = (x_1, \dots, x_{t-1})$ is the preceding context. Equation (2.1) says the model only needs to do one thing: predict the next token given everything before it. This is *next-token prediction*, and it is the only training signal applied to the models studied in this thesis.

Training fits θ by maximizing the likelihood of a large corpus, equivalently minimizing the average negative log-likelihood, known as the cross-entropy loss,

$$\mathcal{L}(\theta) = -\frac{1}{T} \sum_{t=1}^T \log p_{\theta}(x_t | x_{<t}). \quad (2.2)$$

Minimizing Eq. (2.2) rewards the model for placing high probability on the token that actually occurs. No labels beyond the raw text are needed: the target at position t is the observed token x_t , which is why the objective is called *self-supervised*. Optimization proceeds by stochastic gradient descent or its variant such as Adam [31], iterating $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}(\theta)$ over minibatches of sequences, with learning rate η .

2.3 The Transformer architecture

Most current language models are based on the *Transformer* architecture [62], and in particular the decoder-only variant [51]. This architecture turns a context into a next-token distribution in three stages. First, each token is mapped to a d -dimensional vector by an embedding matrix $E \in \mathbb{R}^{|\mathcal{V}| \times d}$ with additional positional information added to it. This produces an initial sequence of vectors $h_1^{(0)}, \dots, h_t^{(0)}$. Second, a stack of L identical layers processes these vectors. Each layer both reads from and writes to a shared *residual stream*, adding information rather than completely overwriting it,

$$\tilde{h}^{(\ell)} = h^{(\ell-1)} + \text{Attn}^{(\ell)}(h^{(\ell-1)}), \quad (2.3)$$

$$h^{(\ell)} = \tilde{h}^{(\ell)} + \text{MLP}^{(\ell)}(\tilde{h}^{(\ell)}), \quad (2.4)$$

where MLP is a position-wise feed-forward network and layer normalization is omitted for clarity. Third, the final-layer vector at each position is mapped back to a score for every token in the vocabulary and normalized into a distribution over the next token,

$$p_{\theta}(x_{t+1} | x_{\leq t}) = \text{softmax}(W_U h_t^{(L)}), \quad \text{softmax}(z)_j = \frac{e^{z_j}}{\sum_k e^{z_k}}, \quad (2.5)$$

with unembedding matrix $W_U \in \mathbb{R}^{|\mathcal{V}| \times d}$.

An important module inside each layer is *self-attention*, the operation that lets a token gather information from earlier ones. It works by linearly projecting each token vector into a query, a key, and a value. A token attends to the earlier tokens whose keys best match its query, and reads out a weighted average of their values,

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}} + M\right)V, \quad (2.6)$$

where the rows of Q, K, V hold the per-token queries, keys, and values, respectively, d_k is the key

dimension, and M is a causal mask that sets attention to future positions to $-\infty$. The mask enforces the autoregressive constraint of Eq. (2.1): the prediction at position t can depend only on $x_{\leq t}$. Stacking many such layers, each followed by an MLP, lets the model build increasingly abstract representations of the context. The representations we probe in Section 4.2 are exactly these residual-stream vectors $h^{(\ell)}$, read at the token position of an entity. The next section describes how these vectors can be analyzed.

2.4 Representational geometry

2.4.1 Concepts as directions

The *linear representation hypothesis* states that human-interpretable features are encoded as directions in a model’s activation space, so that semantic relations correspond to geometric ones [48]. The residual stream is the natural place to look, since every layer reads from and writes to it additively, making it a shared channel whose contents accumulate across depth. Recent work extends the hypothesis to structured concepts: Park et al. [47] show that categorical and hierarchical relations among natural concepts appear as systematic linear structure in pre-trained models, with hierarchy expressed through nested subspaces, and Nava and Wyart [42] argue that such geometry emerges from word co-occurrence statistics.

The practical consequence of the hypothesis is that comparatively simple geometric tools are enough to test claims about a model’s conceptual organization. If concepts are directions and related concepts occupy related directions, then distances and angles between activation vectors carry the relevant information.

2.4.2 Similarity between concept representations

Representations are often compared with cosine similarity,

$$\text{sim}(h_i, h_j) = \frac{h_i^\top h_j}{\|h_i\| \|h_j\|}, \quad (2.7)$$

computed between residual-stream vectors $h^{(\ell)}$ read at token positions of the target concept words.

Two properties of Transformer activation spaces are important for designing and interpreting these measurements. First, the geometry is layer-dependent. Early layers stay close to token identity while later layers carry more context-dependent and high-level semantic information [19, 43, 59], which is why we analyze representations from mid-to-late layers (Section 4.2). Second, a representation read at a token position reflects the surrounding context as well as the concept.

Because of this, similarities are typically averaged over several contexts to keep the comparison about the concept rather than about the phrasing that contained it [47].

2.5 Memorization, generalization, and scale

Nothing in Eq. (2.2) tells a model *how* to reduce its loss on any particular regularity in its training data. This section describes the two solutions available to it, and what is known about how the balance between them depends on the size of the model.

2.5.1 Two ways of fitting the same data

Consider a corpus in which individual items are repeatedly associated with attributes, and in which the items fall into groups whose members share many of their attributes. A model can lower its loss on such data in two ways. It can store each item-attribute association separately, which reduces loss on that association and on nothing else. Or, it can represent items in terms of what they share, so that a single representation serves many associations at once. The second solution is a compression: it costs less capacity whenever the data is redundant, and it generalizes as a side effect, since an attribute attached to a shared representation is attached to every item that shares it. Generalization of the kind studied in this thesis is a consequence of the second solution.

The architecture allows either of the solutions. Sufficiently large networks can fit arbitrary, even randomly assigned, labels, so the capacity to memorize is always present and the question is only which solution the optimization finds [66]. Let us note that the two are not mutually exclusive: in some settings, a memorizing solution is found first and a generalizing one emerges much later in training, long after the training loss has saturated [50]. Feldman [15] makes the further point that memorization is not a failure mode, since memorizing rare examples can be necessary for optimal generalization when the data distribution has a long tail. Memorization and generalization are therefore best understood as competing uses of a finite resource rather than as good and bad behaviors.

2.5.2 The effect of scale

The dominant expectation in modern deep learning is that larger models are better, and the scaling-law literature supports this for the training objective itself: test loss decreases predictably as a power law in the number of parameters, the number of training tokens, and the compute budget [25], with the compute-optimal balance between model size and data characterized by Hoffmann et al. [23]. These laws describe aggregate loss, however, and not the solution by which a model achieves it.

Models with the same loss on data similar to what they were trained on can behave very differently on anything far from their training distribution [14].

What is known about memorization suggests that the balance does shift with scale. Larger models memorize more of their training data, and the number of memorized examples increases with the number of times they are repeated [9, 61].

2.5.3 In-context learning

The description so far concerns knowledge stored in θ and acquired by gradient descent. However, language models also have a second route to acquiring new behavior: *in-context learning* (ICL), the ability to infer a task from examples placed in the prompt and apply it to a new query, with no weight update at all [8]. Concretely, one forms a prompt by concatenating k demonstrations and a query,

$$\text{prompt} = (u_1, v_1, u_2, v_2, \dots, u_k, v_k, u_{k+1}), \quad (2.8)$$

and gets the model’s prediction for the missing v_{k+1} from $p_\theta(\cdot \mid \text{prompt})$ using the same next-token procedure as Eq. (2.1).

What is interesting about ICL is that performance often predictably improves with the number of shots k , as if the model gradually infers the underlying rule from the demonstrations [8]. Explanations range from implicit Bayesian inference over latent tasks [65] to specific circuits such as *induction heads*, which are attention heads that locate an earlier occurrence of the current token and copy what followed it [45]. In practice, demonstrations serve partly to specify the format and label space of the task rather than to teach the mapping completely [37].

The distinction between the two routes matters for this thesis. The knowledge held in θ is *in-weights*: the model must generalize in a single forward pass from what pre-training left in its parameters, with no examples in the prompt. The knowledge supplied in the prompt is *in-context*. The two are not independent: the capacity for in-context learning is itself acquired during pre-training, and the balance between in-weights and in-context strategies is shaped by the statistics of the training distribution, with skewed, bursty data favoring in-context solutions [10]. This thesis measures both, and Section 4.6 examines what happens when context contradicts knowledge stored in weights.

2.6 From token probabilities to property induction

Since a language model can be reduced to Eq. (2.1), everything we may want to measure can be written as a product of next-token probabilities. The central quantity in this thesis, the probability

the model assigns to a property phrase given an entity, is

$$p_{\theta}(\text{property} \mid \text{entity}, \text{context}) = \prod_t p_{\theta}(w_t \mid w_{<t}, \text{entity}, \text{context}), \quad (2.9)$$

where $w_1, w_2, \dots$ are the tokens of the property phrase. A model that has learned that an entity belongs to a category, and that the category carries a given property, will place high probability on that property's tokens even when the specific entity-property pairing never appeared in training. Tracking how this probability evolves over training, and how closely it follows the latent taxonomy, is what lets us study category-based property induction directly from the model's own distribution.

Chapter 3

Experimental Setup

Our experimental setup, as described in this chapter, has three primary goals. First, it must provide a clean test of property induction and taxonomy learning with minimal impact of unaccounted confounders. Our synthetic taxonomies and evaluation are designed to address this. Second, the models and training procedures should remain close to real-world LLM pre-training. We therefore choose openly available model families with released intermediate checkpoints, continue training them on the same corpora they were originally pre-trained on, and keep next-token prediction as the only objective. The synthetic taxonomy is mixed into that stream as a small fraction of the tokens rather than replacing it, so the models stay close to the regime they were trained in. Third, the setup has to let us observe induction as it develops and not only whether it is present at the end. We therefore inject the taxonomy throughout continued pre-training, evaluate at regular intervals, and repeat every run across several random seeds and base checkpoints.

Satisfying these goals at once faces practical difficulties: full control pushes towards artificial data, realism pushes towards natural corpora, and developmental measurement requires enough repeated runs while keeping the experiments affordable. The remainder of this chapter describes how we resolve that tension. Section 3.1 introduces the model families and how we continue pre-training them. Section 3.2 describes the corpora, the construction of the synthetic taxonomy, and how the two are mixed. Section 3.3 defines what counts as correct induction and the measures we use to quantify it.

3.1 Models

We validate our findings across two model families and a range of sizes. The most important hyperparameters and settings used for their continued pre-training are reported in Table 3.1.

Table 3.1: Hyperparameters used for continued pre-training with injected synthetic taxonomy data. Learning rates are loaded from the optimizer checkpoints and are therefore slightly below the reported values due to their original (global) cosine schedules. A constant learning rate (LR) scheduler was chosen as it approximates the global cosine schedule well on our small training window of 10k training steps.

Hyperparameter	Pythia	OLMo 2
Optimization		
OPTIMIZER	AdamW	AdamW
LEARNING RATE	$\approx 3 \times 10^{-4}$	$\approx 4 \times 10^{-4}$
LR SCHEDULER	constant	constant
WARMUP STEPS	0	0
ADAM β_1	0.9	0.9
ADAM β_2	0.95	0.95
WEIGHT DECAY	0.1	0.1
MAX GRADIENT NORM	1.0	1.0
TRAINING STEPS	10,000	10,000
Batch composition		
PER-DEVICE BATCH SIZE	2	2
GRADIENT ACCUMULATION STEPS	16	16
GLOBAL BATCH SIZE (SEQUENCES)	32	32
SEQUENCE LENGTH	2,048	4,096
GLOBAL BATCH SIZE (TOKENS)	65,536	131,072
Precision and efficiency		
PRECISION	bfloat16	bfloat16
GRADIENT CHECKPOINTING	no	no
LOGGING INTERVAL (STEPS)	10	10

3.1.1 Pythia

We use the Pythia suite [7], a set of decoder-only transformers (70M, 160M, 410M, 1B, and 2.8B parameters) trained with an identical data order on the Pile [17], with public intermediate checkpoints spanning the 143k-step training run. The shared data order and dense checkpointing make Pythia well suited to developmental analysis. We continue pre-training from a released checkpoint, resuming the original data stream while mixing in our synthetic taxonomy (Section 3.2). Unless stated otherwise, developmental analyses resume the optimizer state and start from the 50k/143k checkpoint. This allows us to study realistic intermediate stages of pre-training when naturally occurring taxonomic data could be seen by the model for the first time (*e.g.*, in multi-epoch training).

3.1.2 OLMo 2

We complement Pythia with 1B and 7B OLMo 2 [44], trained for 1.9M steps on OLMo-mix-1024. OLMo 2 differs from Pythia in architecture, tokenizer, and data distribution, so it tests whether our findings are family-specific. As with Pythia, we resume pre-training with the next-token objective from a released checkpoint (800k/1.9M) and inject the synthetic taxonomy into the data stream.

3.2 Training Data

Studying induction *as it develops* requires two things at once: a data distribution close to real pre-training, and complete control over which entities co-occur with which properties. Real corpora provide the first but not the second; purely synthetic training provides the second but not the first. We combine them by mixing a small, fully controlled synthetic taxonomy into an otherwise unchanged pre-training stream.

3.2.1 Synthetic taxonomy of concepts

We define a balanced binary tree of depth D over 2^D leaf entities. Each internal node carries a set of *shared properties*, and an entity inherits the properties of all its ancestors, so two entities that share a deeper node share more properties. Unless stated otherwise, we set $D = 3$ in all experiments reported in this thesis, giving 8 leaf entities, and each node carries two shared properties. Every entity therefore inherits six shared properties in total, two from each of the three levels above it, and any two entities agree on exactly as many of them as the depth of their lowest common ancestor allows. We provide an example taxonomy in Figure 3.1.

Apart from the *shared* properties at internal nodes, which define the taxonomy, we also introduce a *probe* property P and *control* properties $C_i \in C$. The probe property is trained on one member of a category but held out from its same-category sibling, so that an increased probability of P for a sibling measures within-category induction. The control properties, on the other hand, are trained only on entities in other categories and serve as decoys for across-category induction (the control property subscript i refers to the entity index). Probe and control properties are therefore unique to individual entities, and some entities have neither (*e.g.*, *rufer* in Figure 3.1), allowing us to evaluate the model’s inductive generalization (Section 3.3).

Entity names and properties are pseudowords produced by the Wuggy generator [29], which derives pronounceable non-words from English (entity names such as *hodger*, *pirer*, and *cosel*, with property predicates such as *gruffres branely* and *sabbles chiedy*). Pseudowords guarantee that the model has no prior association for any entity or property, giving us full control over which entities co-occur with which properties, and how they are organized in the ground-truth taxonomy.

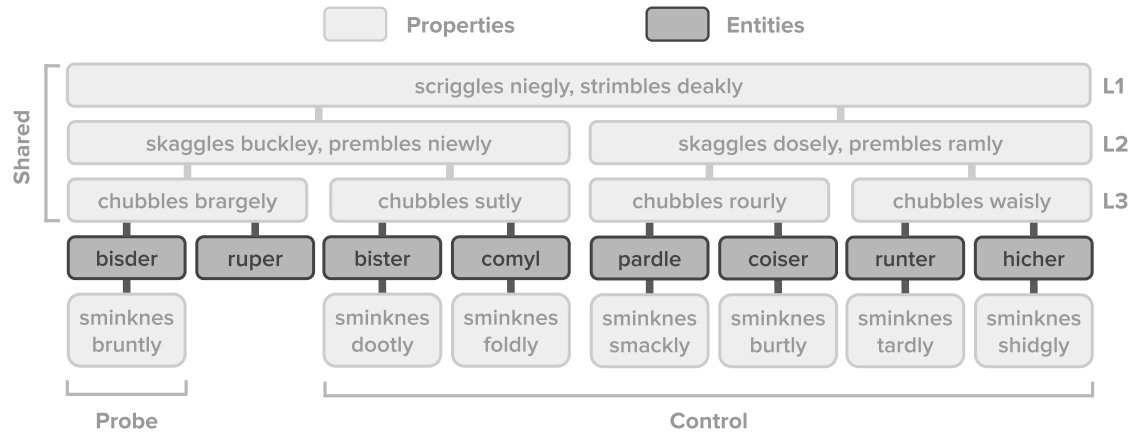


Figure 3.1: Schematic of a synthetic hierarchical taxonomy of depth $D = 3$ with generated pseudoword entities and properties. L1, L2, and L3 denote individual levels of the tree. Some shared properties are omitted for presentation purposes.

Each random seed resamples the entire vocabulary, both the entity names and the properties, as well as their assignment to positions in the tree. Training models across three different random seeds therefore evaluates property induction with three different vocabularies and corresponding taxonomies. This validates that our results do not depend on any particular set of pseudowords.

We render the taxonomy into text by sampling natural-language sentences that state pairs of entities and properties. Using a single fixed phrasing would let the model solve the task by matching one surface pattern, so we vary the sentence template while holding the underlying statements constant. Each training sentence is drawn uniformly from a pool of 30 templates, listed in Box 3.1. The pool is split into three groups that differ in how the property is realized. Direct present-tense frames take the property as a finite verb phrase, as in *hodger gruffres branely*. Indirect frames embed the property under an infinitive or a raising construction, so the property appears in its bare form, as in *pirer tends to gruffre branely*. Past-tense frames take the property in the past, as in *colel gruffred branely*.

During evaluation, we score the probability of a property completion given a prefix, averaged over the 12 templates in Box 3.2. Of these, 2 also appear in the training pool, and the remaining 10 are never seen during training. The held-out majority is what rules out the most immediate alternative explanation for our results, namely that the model has learned to complete a specific surface pattern rather than to project a property onto an entity. The evaluation templates follow the same three groups as the training pool, so the evaluation covers the same range of constructions the model was trained on, and averaging over them prevents any single phrasing from driving the measurement. Templates that end in a particle, such as *often tends to* or *will*, require the bare property form, and the completion is scored taking this requirement into account.

Box 3.1: Training templates

Direct, present tense. The property is realized as a finite verb phrase.

- {entity} {property}.
- It is known that {entity} {property}.
- Research confirms that {entity} {property}.
- Observations show that {entity} {property}.
- According to records, {entity} {property}.
- One key characteristic: {entity} {property}.
- Studies indicate that {entity} {property}.
- It has been documented that {entity} {property}.
- By all accounts, {entity} {property}.
- As widely reported, {entity} {property}.

Indirect. The property is embedded under an infinitive or raising construction and appears in its bare form.

- Based on extensive observation, {entity} has been found to {property}.
- Experts have long noted that {entity} tends to {property}.
- When studied closely, {entity} was observed to {property}.
- Historical records indicate that {entity} is known to {property}.
- In documented cases, {entity} has been shown to {property}.
- Careful analysis has shown that {entity} appears to {property}.
- Those familiar with {entity} know it is prone to {property}.
- It is not uncommon for {entity} to {property}.
- On closer inspection, {entity} turns out to {property}.
- Researchers have come to expect {entity} to {property}.

Direct, past tense. The property is realized in the past tense.

- In all previous iterations, {entity} {property}.
- Initial tests confirmed that {entity} {property}.
- Historically, it was found that {entity} {property}.
- According to the ancient logs, {entity} {property}.
- It was widely believed that {entity} {property}.
- Every time the team checked, {entity} {property}.
- The final report stated that {entity} {property}.
- Until the recent update, {entity} {property}.
- In past surveys, {entity} {property}.
- Observers were surprised when {entity} {property}.

Box 3.2: Evaluation templates

Direct, present tense.

- {entity} {property}.
- It is known that {entity} {property}.
- A defining trait is that {entity} {property}.
- It is a well-established fact that {entity} {property}.

Indirect. These end in a particle and take the bare property form.

- {entity} often tends to {property}.
- Under typical conditions, {entity} will {property}.
- {entity} has been observed to {property}.
- Historical records show that {entity} tends to {property}.

Direct, past tense.

- In earlier observations, {entity} {property}.
- Back then, {entity} {property}.
- It was found that {entity} {property}.
- In the past, {entity} {property}.

3.2.2 Data mixing

Each training batch contains 32 sequences of 2,048 (Pythia) or 4,096 (OLMo 2) tokens. For the first 5,000 of the total 10,000 steps of continued pre-training, we inject synthetic facts from our taxonomy into each training batch. In particular, we inject (1) one sequence per depth level in the taxonomic tree, where each sequence contains all entities and properties at that level, and (2) one sequence with the probe and control properties. For a balanced taxonomic tree of depth $D = 3$, this results in training batches composed of 28 sequences from the background pre-training corpus and 4 sequences with synthetic facts. We pad sequences shorter than the context length with a padding token that we exclude when calculating the loss.

These injections ensure that the synthetic signal is a relatively small share of tokens ($\approx 2\text{--}4\%$ after padding), as a rare but consistent fact would be in real data. While the resulting frequency is not calibrated to any measured rate in natural corpora, we believe this design is a good initial step toward approximating real-world pre-training while still allowing controlled studies of how data organization shapes generalization. We provide a comparison to training with lower ratios of synthetic data and the original batch sizes in Section 4.3.

3.3 Evaluation

We call induction *correct* when it follows the latent taxonomy: a novel property observed for one entity should transfer to entities that share its deepest category, and less to more distant ones. We measure transfer directly from the model’s next-token distribution. For a target entity and a prefix template, we score $\log p(\text{property} \mid \text{entity})^2$ for several candidate completions and average over a set of evaluation templates (Box 3.2). We track three quantities:

- *Memorization*: the probability of a probe property P for an entity that was trained on it (e.g., *bisder sminknes bruntly* in Figure 3.1). This can be viewed as a reference ceiling on what the model can express.
- *Induction within category*: the probability of a probe property P for a same-category sibling that was *not* trained on it (e.g., *rupe sminknes bruntly* in Figure 3.1).
- *Induction across categories*: the average probability of a control property $C_i \in C$ for the same target entity as the induction-within-category measurement, where the C_i belong to different categories.

Correct taxonomic induction means within-category > across-category. We summarize the gap as the *induction advantage*, the difference in log-probability between within- and across-category induction:

$$\text{induction advantage} := \frac{1}{|F|} \sum_{f \in F} \left(\log p(P \mid f(E)) - \frac{1}{|C|} \sum_{C_i \in C} \log p(C_i \mid f(E)) \right). \quad (3.1)$$

In the definition above, P and $C_i \in C$ correspond to the probe and control properties, respectively; E refers to the target entity, and $f \in F$ denotes the prompt templates. The induction advantage for entity E that was trained on neither the probe nor the control properties is positive exactly when the model generalizes along the taxonomy, and we use it to compare across training steps, model sizes, and checkpoints.

3.3.1 Constraining the completion space

The three quantities above are defined over property completions, but comparing whole property phrases directly turns out to be unreliable. Each property consists of a verb and an adverb. Rather than sharing nothing between categories, we share the property verb across categories and compare only the probabilities of the token-length-matched adverbs that follow it. For example, instead of

²For readability, we write $\log p(\text{property} \mid \text{entity})$ throughout, omitting the prompt template and the surrounding text that the probability is also conditioned on.

comparing the probability of *gruffres branely* in *hodger gruffres branely* against that of *crambles peedly* in *hodger crambles peedly*, we share the root verb *gruffre* between categories and compare the completions *branely* and *peedly* after the common prefix *hodger gruffres*.

This design serves two purposes. First, it constrains the space of plausible continuations, which sharpens the contrast between within-category and across-category induction, because both candidates are grammatically and distributionally supported at the position being scored. Second, it removes residual associations for entity names that resemble real words. For instance, *cokol*, produced by the Wuggy generator, is close to the Czech word for falcon, *sokol*. In our initial experiments, some entity names produced much higher entropy in the next-token distribution, which made results unstable across random seeds. Prefilling the property verb largely resolved this issue.

Chapter 4

Experiments

This chapter presents our experimental results, addressing the four research questions from Chapter 1. The outline is as follows. We begin with the existence question **RQ1**: Section 4.1 asks whether taxonomically aligned induction appears at all under next-token prediction, and whether it depends on the taxonomic structure of the training data or merely on the presence of the injected facts or some predispositions of the used models. Section 4.2 then investigates what changes inside the model support that behavior, testing whether the latent taxonomy becomes visible in the geometry of entity representations and whether the tree can be recovered from them (**RQ2**). To further verify these results against realistic training conditions, Section 4.3 performs a controlled comparison of batch sizes, validating that our findings are not an artifact of our specific design choices.

In the remaining sections, we study how induction depends on training and on the information available in the prompt at inference time. In particular, Section 4.4 varies the amount of pre-training that precedes the injected taxonomy and the size of the model (**RQ3**), and finds that these two pull in opposite directions. Section 4.5 then studies what happens when a taxonomy the model has already learned must be replaced by a conflicting one, which tests how deeply the earlier commitment is held by the LMs. Finally, Section 4.6 turns to the interaction between the two routes to generalization described in Section 2.5.3, asking whether a taxonomy learned in the weights can be applied to properties supplied in the prompt, and which of the two wins when they disagree (**RQ4**).

Unless stated otherwise, all experiments are repeated across three random seeds, each of which resamples the entire pseudoword vocabulary and its assignment to the tree, and all shaded regions and error bars report the standard error of the mean across those seeds.

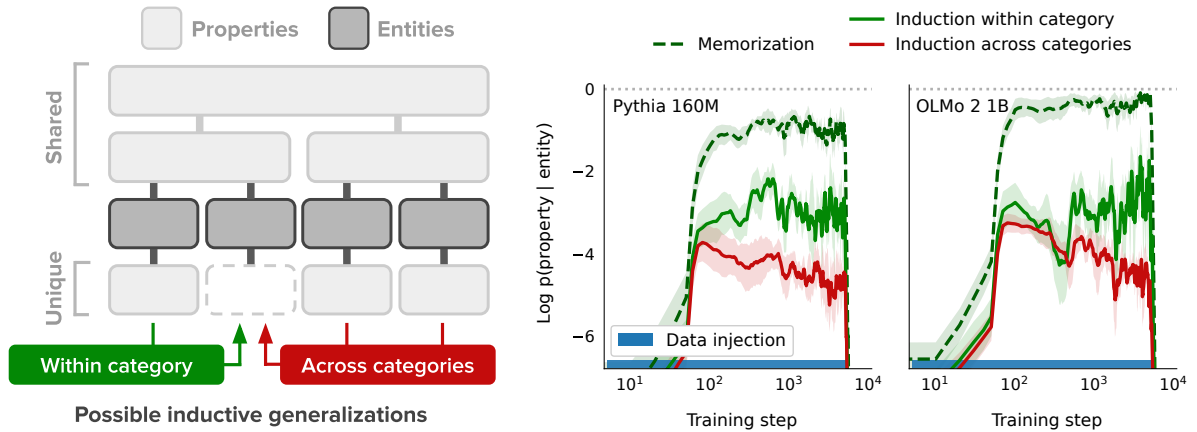


Figure 4.1: Left: Simplified taxonomy schematic. Right: $\log p(\text{property} \mid \text{entity})$ over training for Pythia 160M and OLMo 2 1B, showing memorization (dashed) > induction within category (green) > induction across categories (red). This shows that LMs develop taxonomically aligned inductive generalization. Shaded regions indicate the standard error of the mean across three random seeds.

4.1 LMs inductively generalize along the latent taxonomy

In traditional pre-training, LMs are trained to predict the next token. Nothing in that objective asks them to build categories or tell them which entities belong together. If a model observes that an entity has a certain property, the cheapest way to reduce its loss is to store that single fact. Therefore, we begin by asking whether taxonomically aligned induction appears at all under ordinary pre-training, or whether it needs the kind of explicit supervision used in earlier work [38].

To answer this question, we track the memorization and induction within and between categories over continued pre-training of Pythia 160M and OLMo 2 1B. As Figure 4.1 shows, all three quantities rise as the taxonomy is learned, but they separate from each other as training progresses: memorization is highest, induction within category sits well above induction across categories (*i.e.*, *correct induction*), and the ordering persists for the rest of training. The effect appears in both families, so it is unlikely to be an artifact of a particular architecture or corpus.

Figure 4.1 also sheds light on the development of property induction: the correct induction grows over continued pre-training, so more exposure to the taxonomy yields stronger taxonomic generalization, not merely more memorization. At the same time, however, generalization along the latent taxonomy is forgotten once the evidence for the taxonomy is removed from the training signal (Figure 4.2). In particular, when we stop injecting the synthetic facts at step 5,000 and continue training only on the background corpus, memorization as well as correct induction decays within a few hundred steps. Taxonomic structure is therefore maintained only while the data continues to support it. Chang et al. [11] report the same pattern for individual injected facts during pre-training, and our result extends it from the recall of single facts to the taxonomic organization that supports generalizing them.

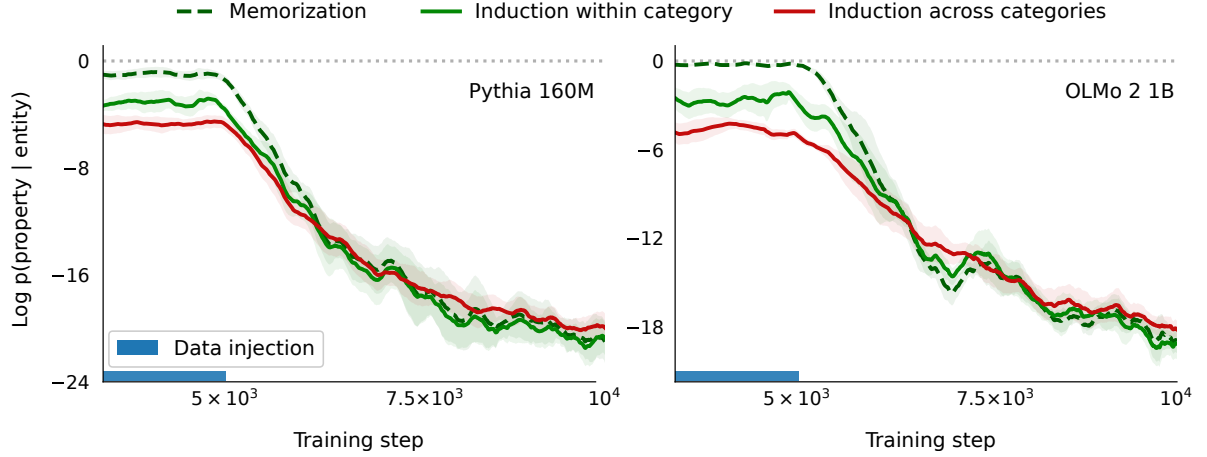


Figure 4.2: $\log p(\text{property} | \text{entity})$ over the last 6,000 continued pre-training steps of Pythia 160M (left) and OLMo 2 1B (right). Shaded regions indicate the standard error of the mean across three random seeds.

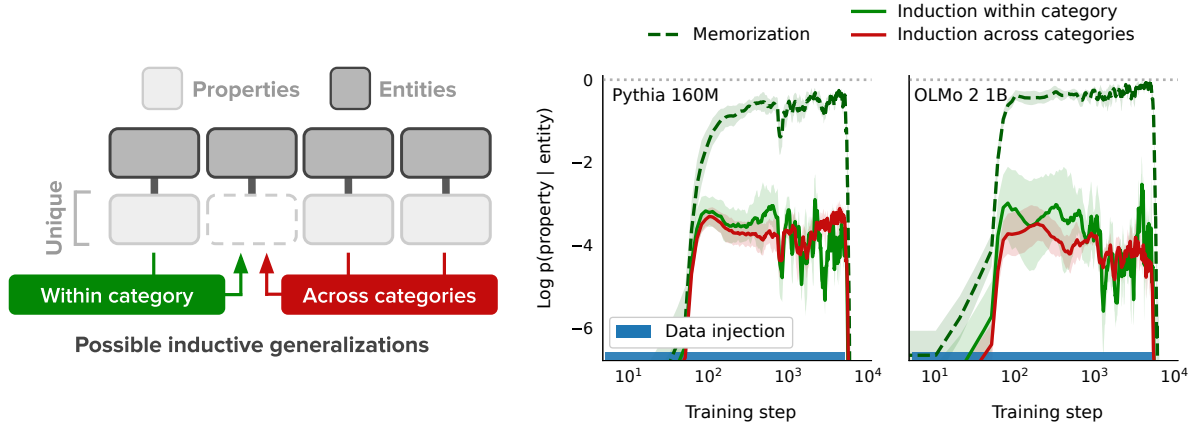


Figure 4.3: Left: Simplified taxonomy schematic for the ablation experiment. Right: $\log p(\text{property} | \text{entity})$ over training for Pythia 160M and OLMo 2 1B, showing memorization (dashed) $>$ induction within category (green) $\approx$ induction across categories (red). This illustrates that removing the shared-property signal removes correct inductive generalization. Shaded regions indicate the standard error of the mean across three random seeds.

Ablation: no shared properties. The correct induction could still come from the model’s existing predispositions to link certain entity names together rather than from the taxonomy. To rule this out, we remove all taxonomy-related information from the continued pre-training. Specifically, we keep the entities and the probe and control properties but drop the shared ones, so the model has no signal about which entities can be categorized together. Under this ablation, within-category and across-category induction collapse onto each other (Figure 4.3). Memorization is almost unaffected, which confirms that the model still learns the facts it is shown and only loses the ability to project them correctly.

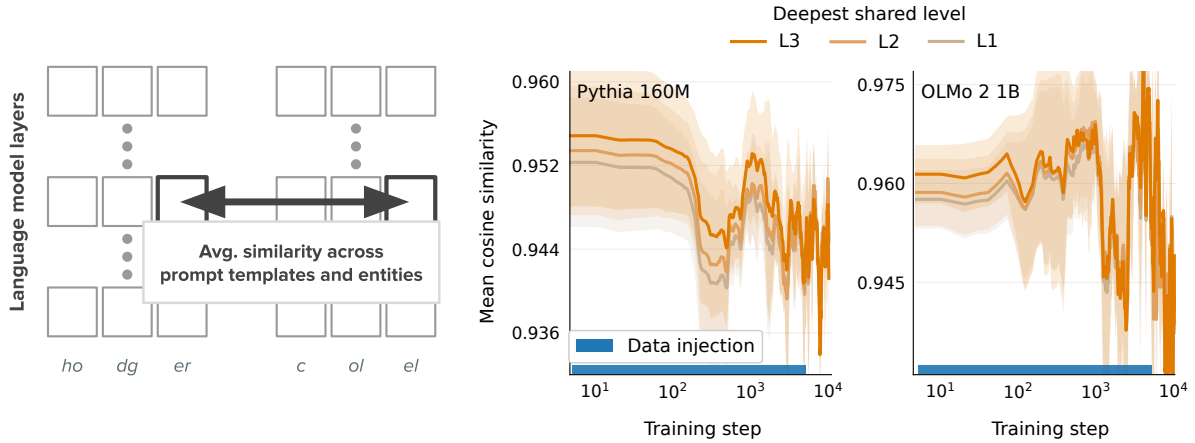


Figure 4.4: Left: Illustration of the similarity computation between entity representations. Right: The progression of mean cosine similarity between representations of entities across the taxonomy, grouped by the deepest shared hierarchy level they share. Shaded regions indicate the standard error of the mean across three random seeds.

Taken together, these results establish the central claim of the thesis. **The latent taxonomic structure of the data is what produces correct property induction.** This matters for deciding what to train a model on: it means that the inductive behavior of a model is a property of how its training data is organized, and it can be created or destroyed by changing that organization.

4.2 The taxonomy is recoverable from entity representations

Where does the correct induction come from? Saxe, McClelland, and Ganguli [57] showed that in deep linear networks, hierarchically organized concepts that share deeper categories become more similar in representation. Whether this carries over to Transformers is not obvious, since they compose many non-linear operations, use attention rather than fixed weights between fixed units, and are trained on natural text rather than on clean one-hot attribute vectors.

To answer where the correct property induction might come from, we take the last-token entity representations from mid-to-late layers of our trained checkpoints (layer 8 out of 12 for Pythia 160M, and layer 8 out of 16 for OLMo 2 1B) and measure the mean pairwise cosine similarity within and between categories, averaged across all evaluation templates. The prediction based on [57] would be that two entities that share a deeper node in the tree should be more similar than two that share only a shallow one. Figure 4.4 shows exactly this. The average similarity increases monotonically with the deepest shared level of the ground-truth taxonomy ($L1 < L2 < L3$, Section 3.2). This result illustrates how the representation space is reorganizing to match a structure that appears nowhere in the training text as an explicit statement.

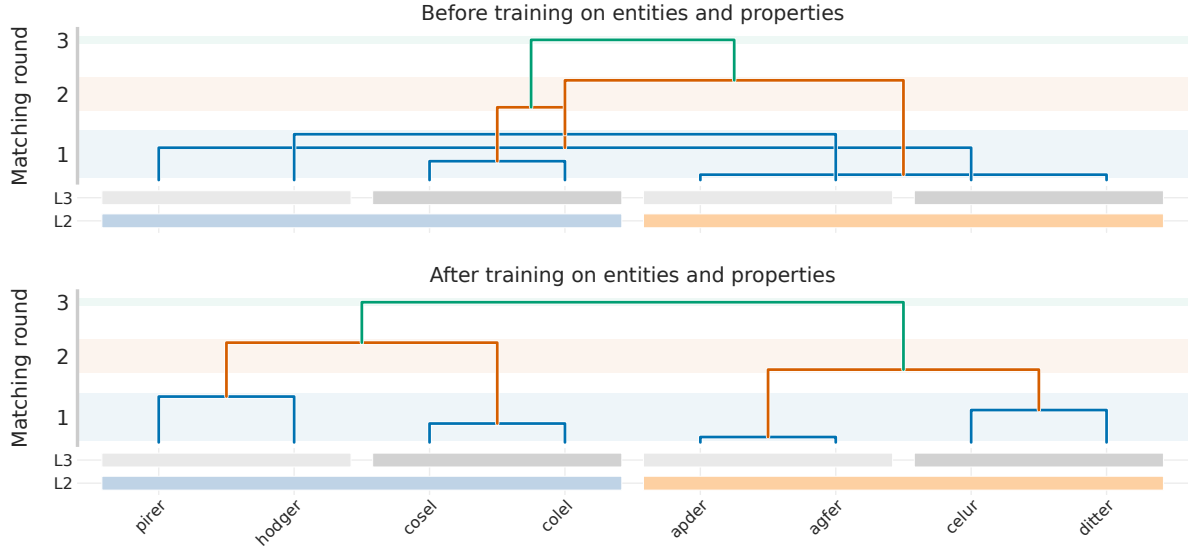


Figure 4.5: Hierarchical recovery of the taxonomy from entity representations of OLMo 2 1B. Dendrograms before vs. after continued pre-training on entities and properties.

Tree recovery. Graded similarity by the deepest shared level is suggestive but not full evidence: it is consistent with a model that has learned only that some entities are similar, but whose internal structure does not mirror the complete taxonomic binary tree. A stronger test would be to assess whether the entire tree, including its branching structure, can be reconstructed from the representations. We use *Matching Affinity Clustering* [20] for this. It starts with all entities as initial clusters. Then, in each round, it partitions current clusters into disjoint pairs by a minimum-weight perfect matching on the complete graph whose edge weights are the average-linkage cosine distances between clusters, and then merges every matched pair. Because each round matches every cluster exactly once, the number of clusters halves per round and the procedure yields a balanced binary tree in $\lceil \log_2 N \rceil$ rounds (for N a power of two), matching the shape of our ground-truth taxonomy. In contrast, standard agglomerative clustering performs $N - 1$ sequential merges, each joining the closest available pair, and generally produces unbalanced trees, which is why we do not use it here.

After the data injection phase of our continued pre-training, we find that the recovered tree matches the ground-truth taxonomy. The recovery is perfect for OLMo 2 1B (Figure 4.5, bottom) and near-perfect for Pythia 160M (Figure 4.6, bottom). The same models before taxonomy injection do not recover it (Figures 4.5 and 4.6, top), so the structure is not an artifact of the clustering algorithm or of how pseudowords happen to be tokenized.

This gives direct empirical support to the representational account of semantic development by Saxe, McClelland, and Ganguli [57], now in the setting of modern Transformer LMs trained on natural text. It also explains the behavior in Section 4.1: correct induction happens because entities that share a category end up close together in representation space, so raising the probability of a property for one entity raises it for its neighbors as a side effect. It also means the taxonomy a model

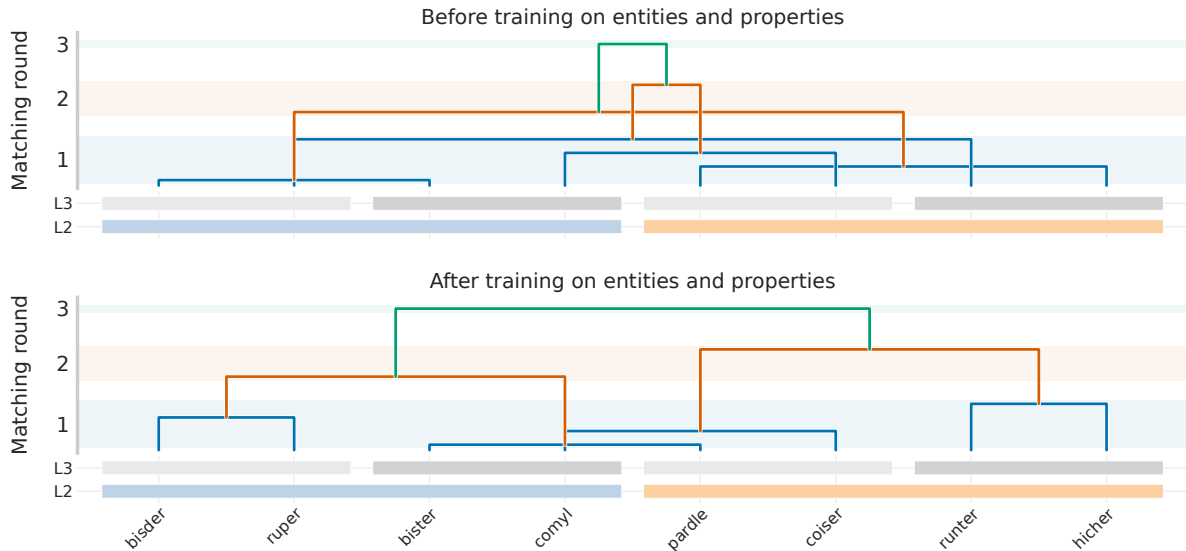


Figure 4.6: Hierarchical recovery of the taxonomy from entity representations of Pythia 160M. Dendrograms before vs. after continued pre-training on entities and properties.

has committed to can be analyzed without looking at its outputs at all. This offers a way to check whether a model has organized a domain the way its users expect before that mismatch shows up in an answer.

4.3 The effect of batch size

For all experiments so far, we use a batch size of 32 sequences, or roughly 65k tokens for Pythia and 130k tokens for OLMo 2. This is much smaller than the 1,024 sequences (2M tokens) and 512 sequences (2M tokens) used in the original Pythia and OLMo 2 training runs, respectively [7, 44]. We chose a smaller batch size due to the available compute budget: continued pre-training has to be repeated across model sizes, checkpoints, seeds, and ablations, and the full batch size would have made that grid unaffordable. However, if our results only held at a batch size we chose for convenience, they would say more about our training setup than about language models. We therefore compare our 32-sequence setting with continued pre-training of Pythia at the original batch size of 1,024 sequences. We hold everything else fixed, including the taxonomy, the number of injected facts into each training batch, the evaluation templates, and the seeds.

As can be seen in Figure 4.7, the ordering of memorization above within-category induction above across-category induction is preserved at the larger batch size, so the qualitative finding does not depend on our choice. The main differences are only in the overall magnitude of all three quantities and in the speed of acquiring the correct induction. In particular, larger batch sizes lead to slower differentiation between within- and across-category induction, and to lower probabilities

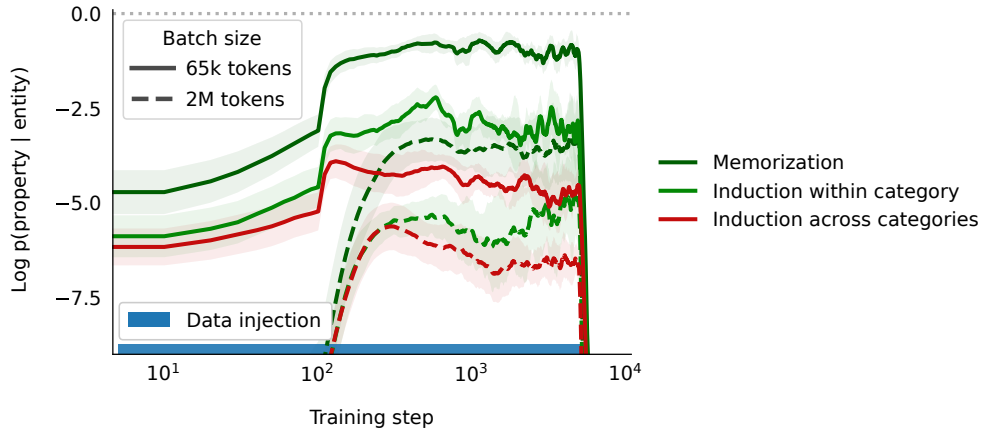


Figure 4.7: Comparison for Pythia 160M between the batch size used in all our experiments (32 sequences; 65k tokens) and the originally reported batch size by Biderman et al. [7] (1024 sequences; 2M tokens). Shaded regions indicate the standard error of the mean across three random seeds.

assigned by the model to our synthetic properties. This can be explained by the fact that the ratio of synthetic entity-property data to the background corpus decreases (≈ 0.06 – 0.13% after padding), leading to lower influence of the synthetic sequences on each training step.

What matters for the rest of the thesis is which parts of the result are robust to this choice. The ordering of the three curves is the claim that later sections rely on, and it is the part we need to be independent of batch size. Absolute values of the induction advantage are more fragile, which is why we compare them only within a fixed batch size and never across the two settings.

4.4 Development and the effect of scale

If induction is learned predictably given the latent structure in the data, then how much a model already knows before it encounters a new taxonomy should matter, and so should the capacity it has to learn. Both are practical questions, since a model that will be adapted to a new domain is always adapted from a certain checkpoint, and practitioners must choose which one and for a model of what size.

The developmental literature suggests that induction in later learning stages might be more taxonomy-aligned, since children generalize more systematically as their existing category knowledge matures [3, 16]. In terms of model scale, the usual expectation in deep learning is that bigger is better, since loss falls predictably with parameters, data, and compute [23, 25], and downstream capability generally follows [8, 64].

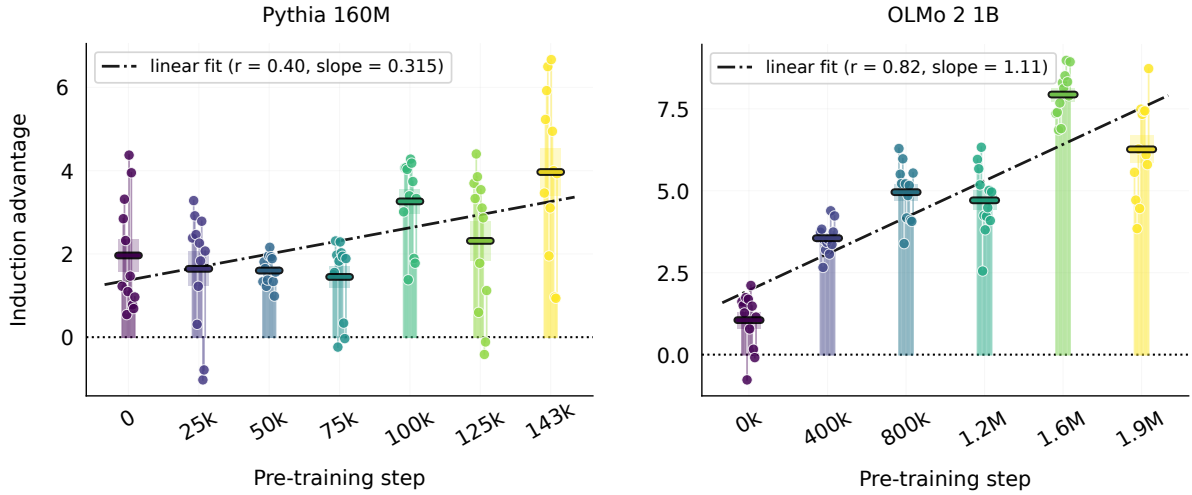


Figure 4.8: More mature checkpoints acquire induction more strongly. Shown is the induction advantage averaged over the last 1,000 steps of synthetic data injection in continued pre-training from various checkpoints (x-axis). Circles show induction advantage for individual evaluation templates, while horizontal lines show the average induction advantage. All reported values are averages from three random seeds.

More mature checkpoints acquire induction more strongly. We repeat the experiment from Figure 4.1 for Pythia 160M checkpoints spanning 0k to 143k pre-training steps, and for OLMo 2 1B checkpoints from 0k to 1.9M steps. In particular, we vary the base checkpoint from which we continue pre-training, holding the taxonomy and vocabularies the same. To quantify the strength of inductive generalization with a single number, we average the induction advantage over the last 1,000 continued pre-training steps of the data injection.

As shown in Figure 4.8, the average induction advantage increases with the base checkpoint’s pre-training step. With the exception of a few checkpoints, the overall trend shows that a model that has learned more general language before the taxonomy is introduced learns to generalize over it more strongly. The capacity to correctly inductively generalize is therefore itself something that pre-training facilitates, and the injected taxonomy lands on more or less prepared model parameters depending on when it arrives.

At a fixed data budget, scale hurts. Scaling the model has the opposite effect. Across Pythia 70M–2.8B and OLMo 2 1B–7B trained on identical injected data, the induction advantage *decreases* with model size (Figure 4.9). The 70M Pythia model and OLMo 2 1B show the strongest correct induction, while larger models exhibit a relatively weak one, most clearly at the endpoints.

We understand this as a trade-off between memorization and generalization. On one hand, larger models fit the injected facts fast enough to memorize them directly, which weakens the pressure to generalize along the taxonomy to minimize loss on the shared properties. On the

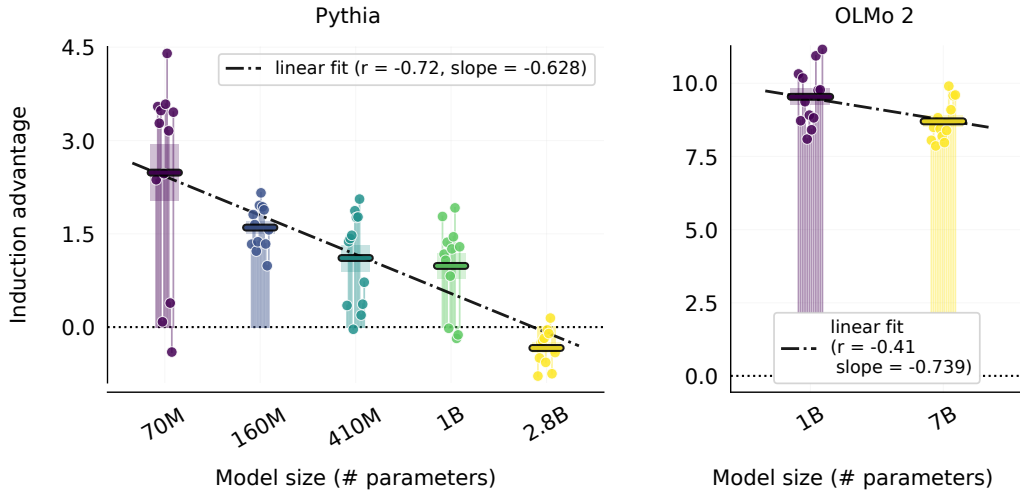


Figure 4.9: At a fixed data budget, scale hurts. Shown is the induction advantage averaged over the last 1,000 data injection steps vs. model size. Circles indicate induction advantage for individual evaluation templates, while horizontal lines show the average induction advantage. All reported values are averages from three random seeds.

other hand, a smaller model cannot store every entity and property pair separately fast enough, so representing the entities in terms of the categories they share is the cheaper solution, and that shared representation is what produces induction.

To further support the explanation that given the same data, scale buys memorization at the expense of correct induction, we compare the speed at which each is acquired. We perform this analysis on the Pythia family, as it provides a denser span of model scales. In Figure 4.10, we plot memorization as defined in Section 3.3 over training for each model size (left), along with the induction advantage averaged over the first 1,000 continued pre-training steps (right). The ordering of the curves on the left confirms that larger models memorize faster, while the downward trend on the right shows that they inductively generalize along the taxonomy more slowly.

These two findings go in opposite directions and complicate a common assumption. More pre-training improves property induction, but more parameters at a fixed data budget do not. Practitioners injecting new domain knowledge into larger models should expect them to more readily recall rather than inductively generalize, unless the data gives the taxonomy enough support to make generalizing worthwhile. Given a fixed dataset, scale alone does not reliably deliver stronger inductive behavior, and may work against it.

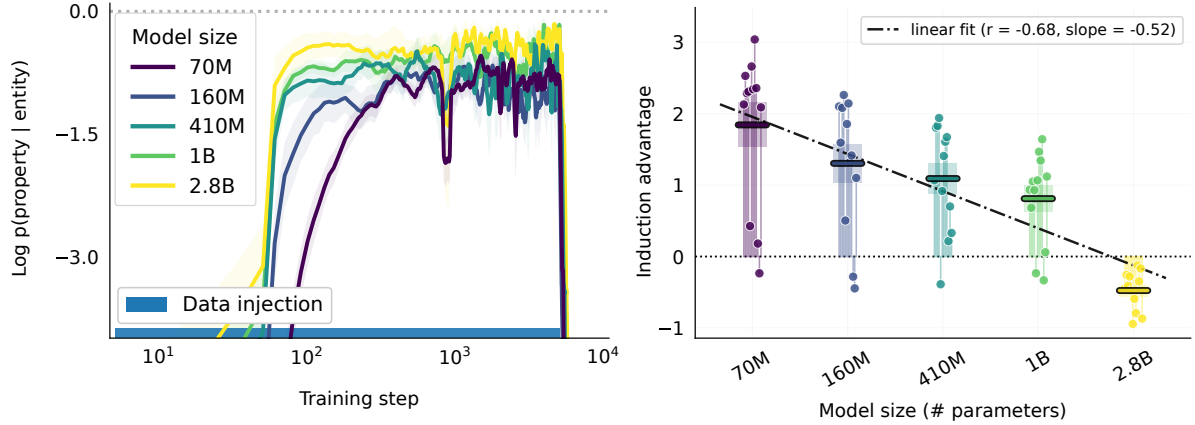


Figure 4.10: Memorization and early induction advantage (steps 0–1,000) vs. model size. Model scaling increases the speed of memorization while reducing correct inductive generalization. Circles indicate induction advantage for individual evaluation templates, while horizontal lines show the average induction advantage. All reported values are averages from three random seeds.

4.5 Flexibility and interference across taxonomies

Category structure in the real world may not always be fixed. Fields reorganize, companies restructure their products and services, and any model will at some point need to update its knowledge regarding a taxonomy it already learned. Learning a taxonomy from almost a blank slate, as described in the previous sections, may therefore be only the first step. The harder but realistic case is the revision, where the model must reorganize the taxonomic structure already acquired in its weights. Beyond practical importance, the study of revision may also deepen our understanding of how models learn such structures in the first place. In particular, if the taxonomy is stored as a set of independent facts, it might be easy to overwrite. But if it has reshaped the representation space, as Section 4.2 suggests, we might see it resist. To test this, we first train language models on taxonomy T_1 , and then continue training on a conflicting taxonomy T_2 that reuses the same entity names but regroups them into different categories (see Figure 4.11, top).

The results in Figure 4.11 demonstrate two things. First, correct induction under T_1 is gradually forgotten once T_2 training begins (dashed lines on the left). Second, the correct induction within the new taxonomy T_2 is weaker than the induction a fresh model would acquire for the same taxonomy (Figure 4.11, right). This combination means that revising leads to the model ending up in the worst of both positions: it loses what it acquired, and it learns the replacement less well than a model starting from a clean slate. We interpret this as interference: a prior taxonomic commitment makes a conflicting one harder to learn, because the entities carry a residue of their earlier categorization that the new training has to work against.

For practice, this argues against assuming that a model’s internal taxonomy can be updated by training on the corrected version. The old structure goes away, so the model will not keep giving the

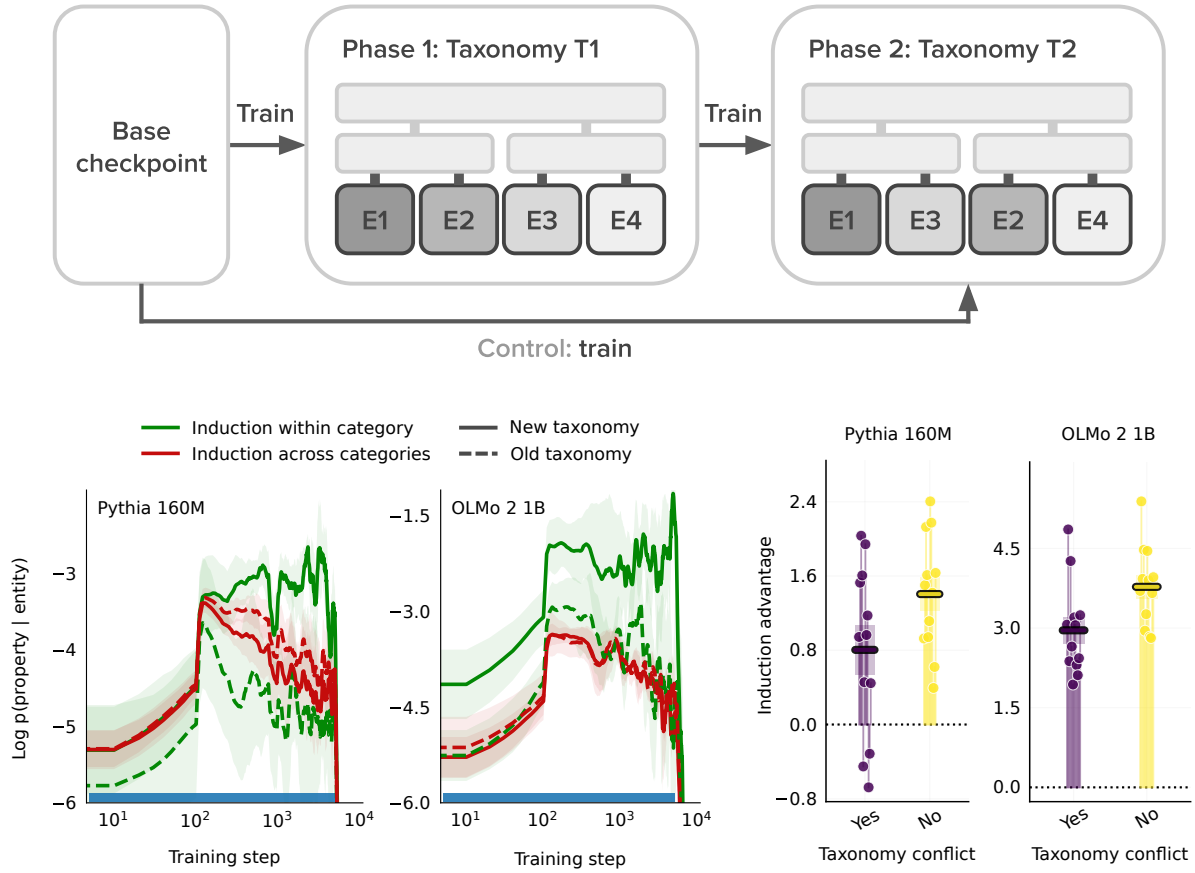


Figure 4.11: Prior taxonomic knowledge is forgotten under, and interferes with, a conflicting taxonomy. Top: Schematic of the experiment. *Control* denotes standard training without taxonomy conflict (Taxonomy conflict: *No*). Bottom: Induction within vs. across categories for the two-phase training (Taxonomy conflict: *Yes*).

old answers, but the new structure might not be learned as cleanly as it would have been in the first place. Therefore, where the correct grouping is known in advance, it is best to train on it from the start rather than to fix it later.

4.6 In-weights taxonomy shapes in-context induction

All results presented so far concern knowledge the model acquired through training. In deployment, however, most of the information a model is given arrives in the prompt instead. Training and ICL described in Chapter 2 therefore meet in every real interaction, and how they combine determines what a user can actually get a model to do. We ask two questions here: (1) does a taxonomy learned in the weights help the model reason about properties it only sees in context, and (2) what happens when the prompt and the weights disagree?

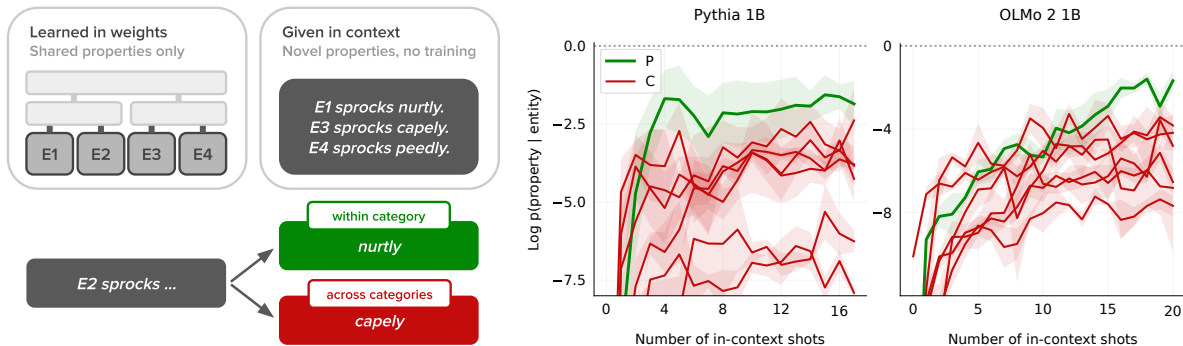


Figure 4.12: Learning a taxonomy in weights enables in-context inductive generalization. Left: Schematic of the experimental setup. Right: The strength of inductive generalization of probe property P (green line) vs. control properties C_i (red lines) to the target entity E_2 . One shot denotes a list of the probe property and all the control properties. The maximum number of shots is set by each model’s context window. Each sentence uses a randomly sampled evaluation template (Box 3.2), and experiments are repeated across five random seeds.

Learning a taxonomy enables in-context induction of novel properties. We train Pythia 1B³ and OLMo 2 1B only on the shared properties of the taxonomy, so they learn the category structure but never see the probe or control properties. We find that a few in-context examples that assign properties to entities are then enough for the models to generalize inductively. In particular, as we repeat the set of sentences containing the probe and control properties (*in-context shot*), the within-category completion of the probe property P is preferred over the control properties C_i (Figure 4.12). This shows that acquiring taxonomic structure in the weights enables models to apply it to novel, in-context properties.

A taxonomy in the weights is therefore not limited to the properties it was learned from. It acts as a reusable structure that the model applies to new properties supplied at inference time. What pre-training provides is the organization that lets new information be projected correctly, instead of a set of memorized rules.

Context overrides weights under conflicting taxonomies. In chatbot applications, users might supply categorizations that contradict pre-training. For example, an internal document used in pre-training might organize products by supply chain, while a user could try to categorize them in the prompt by function. Whether such corrections are effective, or whether the model reverts to the taxonomy it learned during pre-training and keeps generalizing along it, sets a limit on how far a model can be steered by the context it is given.

We test this conflict by first training Pythia 1B and OLMo 2 1B so that their weights favor stronger inductive generalization of the probe property P from entity E_1 to entity E_2 (Figure 4.13, *in-weights*

³We use Pythia 1B rather than the 160M model because we found the smaller model unable to copy a property from the context onto a repeated entity, which is a prerequisite for the in-context form of this task.

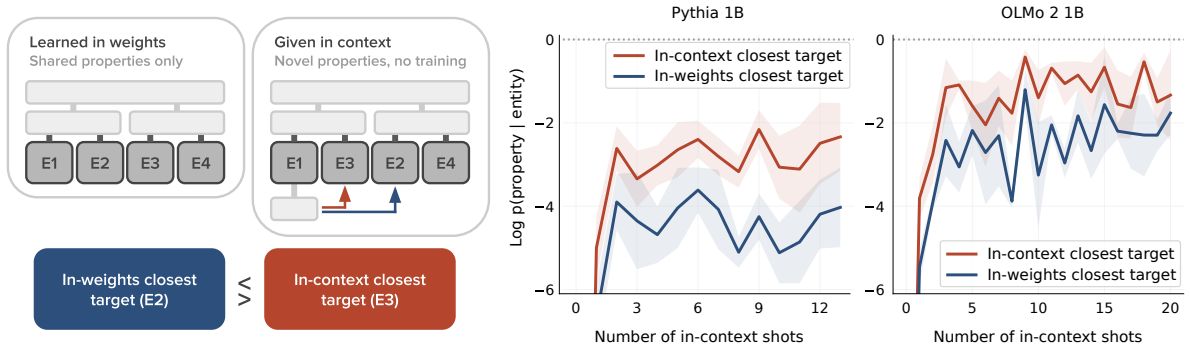


Figure 4.13: When context and weights disagree, the model generalizes according to the context. Left: Schematic of the experimental setup. Right: The strength of inductive generalization of the probe property P to the in-context closest target E3 vs. to the in-weights closest target E2. One shot denotes a list of all the properties in the in-context taxonomy. The maximum number of shots is set by each model’s context window. Each sentence uses a randomly sampled evaluation template (Box 3.2), and experiments are repeated across five random seeds.

closest target) than to entity E3. Then we evaluate these trained models with in-context shots supporting the opposite grouping (stronger $E1 \rightarrow E3$ generalization than $E1 \rightarrow E2$). We can see that as in-context shots accumulate, inductive generalization to the in-context closest target E3 overtakes generalization to the in-weights target E2, and the model generalizes according to the context taxonomy (Figure 4.13).

This result has two implications. First, it is encouraging for controllability, since a user who supplies enough evidence can redirect how a model generalizes without any retraining, and the correction does not need to be anticipated when the training data is collected. It is less encouraging in the other direction. The same mechanism means that a handful of in-context examples can override a taxonomy the model learned from far more evidence, including cases where the pre-trained grouping was the correct one and the context is misleading. Taken together with Section 4.5, this gives a complete picture of how a taxonomy in a model can be changed. Retraining removes the old structure but is worse at learning its replacement, while context leaves the old in-weights structure and overrides it only for as long as the evidence stays in the prompt.

Chapter 5

Related Work

This thesis is at the intersection of three lines of literature: the cognitive science of category-based induction, which defines the behavior we measure, the study of what language models know about categories, which tells us what to expect from a trained model, and the study of learning dynamics, which tells us how such knowledge might develop.

5.1 Category-based induction in cognitive science

The phenomenon we study was defined and extensively studied in cognitive psychology. Rips [53] first showed that projection of a novel property between categories tracks similarity and typicality instead of logical entailment. Osherson et al. [46] described a set of regularities, including premise typicality, diversity, similarity, monotonicity, and conclusion specificity, and captured them with a single similarity-coverage model. Sloman [60] offered a feature-based alternative explanation requiring no explicit category representation, and later work represented induction as Bayesian inference over structured hypotheses such as taxonomic trees [27, 28]. In parallel, the developmental literature established that the taxonomy supporting these inductive inferences is itself learned, and that it differentiates from coarse to fine over years of exposure [26, 34, 35, 49].

These works give us our definition of correct generalization and the two phenomena we target in Section 3.3. It also shows the constraint human studies operate under: since participants have years of prior experience, the experimenter can guarantee no prior exposure to the property by using blank predicates, but cannot do the same for the categories, which have to be familiar to the participants for the task to make sense to them. The taxonomy participants are using therefore needs to be inferred by the experimenter, and the participants' lifetime exposure to it cannot be quantified. Our setting removes that constraint by making both the property and the entities novel, at the cost of a taxonomy far smaller and more regular than a human conceptual system. Additionally, compared

to cognitive science studies, we do not collect graded verbal judgments and instead interpret token probabilities as the argument strength.

5.2 Property induction in language models

The work of Misra, Rayz, and Ettinger [38] is one of the first bringing the property induction paradigm to language models. They fine-tune a model to give yes/no responses to property statements, evaluate induction in the same format, and find that inductive preferences follow taxonomic information latent in the data. This established that the paradigm transfers to LMs and that graded argument strength can be recovered from a model. The design choice that makes it work, explicit supervision for property verification, is also what separates it from our work. In particular, yes/no fine-tuning is not how general-purpose LMs acquire most of their semantic knowledge. To close this gap, we study pre-training, keeping next-token prediction as the only signal. By doing so, we bring our results closer to standard practice at the cost of a less direct read-out of the judgment.

COMPS [39] broadens these initial findings by using minimal pairs to test property knowledge and inheritance, showing that models are sensitive to the taxonomy but also distractible by unrelated context. More recently, Han et al. [21] compared humans, GPT-3.5, and GPT-4 on the phenomena of Osherson et al. [46] and found broadly human-like patterns. Their evaluation is purely in-context, with premises in the prompt and the judgment produced in a single forward pass. We instead study in-weights induction, where the property is learned through gradient updates and any generalization has to come from what the parameters internalized. The two measure different routes to the same behavior, and Section 4.6 shows what happens when they disagree.

Rodriguez, Mueller, and Misra [54] show that the property inferences of instruction-tuned models are jointly explained by taxonomic relations and pairwise categorical similarity, and do so for model sizes from 1B parameters and above. They evaluate a final checkpoint with a blank property but real entities. Our work differs in that (1) we track development across pre-training rather than a single endpoint, (2) we study processes happening during initial knowledge acquisition in pre-training, instead of fine-tuning for a task, and (3) we use pseudowords for both properties and entities, which removes taxonomic structure that the model’s prior knowledge of the entities could provide.

5.3 Learning dynamics and semantic development

Our developmental analysis is closely tied to the connectionist theory of semantic development [55, 57]. Saxe, McClelland, and Ganguli [57] show that deep linear networks trained on hierarchical category-property matrices admit closed-form learning trajectories, predicting progressive differentiation from coarse to fine categories, and a geometry in which similarity between two items

reflects the depth of their shared category in a latent taxonomy. The linearity and the clean attribute matrices are what make the analysis solvable. However, the predictions have not been tested in deep nonlinear models trained on natural text, which is what this thesis brings. Related accounts of coarse-to-fine learning without a taxonomy come from the distributional simplicity bias, where networks fit low-order statistics before higher-order ones [4, 52], and from work identifying discrete developmental stages in transformer training [24].

Empirical studies of learning dynamics in language models establish an order of acquisition, with syntactic information developing before world knowledge and reasoning [13, 33, 40, 56]. These studies cover far more phenomena than we do and use real linguistic knowledge rather than an injected structure. The limitation this set of works shares is that the target knowledge is already present in the corpus in an unknown quantity and distribution. In turn, an acquisition plot mixes the model’s training dynamics with the statistics of the data. By injecting a taxonomy at a known rate from a known step, we separate the two.

Closest to our setting is work on how factual knowledge is acquired during pre-training. Chang et al. [11] track acquisition and forgetting of individual facts. They treat facts as atomic rather than as inductively related members of a category, which is the distinction our probe and control properties are designed to capture. Zucchet et al. [67] inject a synthetic factual recall task and find distinct learning phases, a loss plateau that occurs during the formation of recall circuits, and a strong dependence of the dynamics on the training data distribution. Their findings support our choice to analyze induction over training rather than at its end. Closely related to this thesis is also the work of Kim et al. [30]. They report that readiness to acquire new knowledge declines over pre-training as the model’s knowledge entropy decays, so that mid-stage checkpoints are the best starting point for injecting new facts. Our finding in Section 4.4 that more mature checkpoints acquire stronger induction does not contradict this, since the two measure different quantities. Specifically, they analyze how easily a new fact is absorbed while we study how strongly an absorbed fact is generalized along a taxonomy. Taken together, this suggests that the checkpoint which best memorizes new information and the one which best generalizes over it do not need to be the same.

5.4 Controlled injection and the reach of a learned fact

The method of injecting synthetic data into pre-training is shared with Allen-Zhu and Li [1]. They use generated biographies to show that knowledge becomes extractable only when sufficiently augmented, and they connect this to whether attributes are linearly encoded at the entity token. Their generalization concerns the format, asking whether a fact learned in one phrasing can be retrieved in another. In contrast, our generalization is between entities, asking whether a fact learned about one entity transfers to a different one. Their capacity results also relate to Section 4.4, where we understand the weakening of induction with scale as larger models having enough room to store facts separately and therefore less pressure to compress.

Several results from prior studies describe how narrow generalization from training can be. Models trained on a relation in one direction often fail in the other [5], and Lampinen et al. [32] show in matched comparisons on controlled synthetic facts that in-context learning generalizes more flexibly than fine-tuning.

5.5 The geometry of concepts

Under the linear representation hypothesis [48], semantic relations show up as geometric ones in activation space. Park et al. [47] formalize this for categorical and hierarchical concepts, representing categories as polytopes and hierarchy as nested subspaces, and validate the account on more than 900 WordNet concepts. Nava and Wyart [42] argue that such geometry follows from word co-occurrence statistics. Both analyze end-of-pre-training checkpoints on natural concepts, which is what allows the claims to be about real conceptual structure at realistic scale. What is unavailable in that setting is the ground truth for which concepts should be more similar, and a quantification of the data required for constructing the taxonomic structure inside models. Our controlled taxonomy provides both, letting us track how taxonomically aligned representations develop and link that change to the behavioral trajectory.

Chapter 6

Conclusion

This thesis asked whether language models develop category-based property induction from next-token prediction alone, and if so, how that ability changes over the course of training. The question matters because induction is the mechanism by which a fact stated about one entity moves a model’s predictions about others. Such generalization is what makes models useful: a model that only reproduces its training data is of little value, while a model that generalizes well fills gaps its training data did not cover. However, a model that generalizes along the wrong category structure fills those same gaps with confident yet incorrect or unsupported claims.

We approached the question by constructing a setting in which the answer could be measured without ambiguity. Instead of testing pre-trained models on real categories, where it is extremely difficult to know what the model was already trained on, we constructed a taxonomy of pseudoword entities and properties. This structure was then injected into the continued pre-training of Pythia and OLMo 2 language models alongside their original corpora. Our setup ensured that every co-occurrence the model saw was one we placed there, so nothing could be attributed to prior knowledge. Additionally, the training signal remained the standard next-token prediction objective, so the mechanism we measured was close to the one that operates in practice.

6.1 Summary of findings

Property induction emerges from next-token prediction alone. Language models trained only to predict the next token come to prefer properties that the latent taxonomy supports. Given a property observed for one entity, they raise its probability for entities in the same category well above entities in other categories, and they do so for entities that were never trained on that property (Section 4.1). The effect holds for both model families we tested, which differ in architecture, tokenizer, and pre-training corpus.

In the ablation experiment (Section 4.1), when the shared properties are removed and there is no signal about which entities belong together, within-category and across-category induction collapse onto each other while memorization is left almost unchanged. This showed that correct induction is a consequence of the latent structure in the data.

The taxonomy is written into the representations. The behavior has a representational explanation (Section 4.2): entities that share a deeper node in the tree become more similar in representation, and the ordering across levels of the taxonomy is monotonic. Furthermore, the full taxonomic tree can be reconstructed from these internals. In particular, Matching Affinity Clustering over entity representations recovers the ground-truth taxonomy after injection, perfectly for OLMo 2 1B and near-perfectly for Pythia 160M, while the same models before injection recover mostly random taxonomic trees. This observation provides direct empirical support for the explanation of semantic development by Saxe, McClelland, and Ganguli [57], which was derived for deep linear networks trained on clean attribute vectors and had not been tested in Transformers trained on natural text. The theory, as well as the empirical results with Transformers in this thesis, suggests that category-based property induction works because entities that share a category get placed close together in a representation space. As a result, raising a property’s probability for one entity raises it for its neighbors as a side effect.

More pre-training helps while more parameters do not. Results in Section 4.4 show that the strength of acquired induction grows with how much pre-training preceded the injection of the taxonomy-related data. A model that has learned more general language before it meets the taxonomy learns to generalize over it more strongly, so the capacity to induce correctly is itself something pre-training facilitates.

Interestingly, scaling the model at a fixed data budget has the opposite effect (Section 4.4). Across Pythia 70M to 2.8B, the strength of correct induction decreases with size. We explain this as a trade-off between memorization and generalization: a large model can afford to store each entity and property pair separately, which removes the pressure to represent entities in terms of what they share. A smaller model cannot afford the separate storage and so it compresses instead. This is the most counterintuitive result in the thesis and the one with the clearest practical importance.

Prior taxonomies interfere with revision. What happens when taxonomies change? Section 4.5 tried to answer this by training a model on one taxonomy and then continuing training on a conflicting one that regroups the same entities. The results showed that the model slowly forgets the old organization while its replacement is learned less well than it would have been from a clean start. This provides further support that taxonomies reshape the representation space. In particular, entity representations carry a residue of their earlier grouping that the new training has to battle against.

In-weights taxonomy enables in-context induction, and context overrides weights. Section 4.6 then studied how prior taxonomic commitments interact with in-context information. To isolate taxonomy learning, it trained language models only on shared properties that support the taxonomy. Then, it presented models with new probe and control properties in the prompt, finding that the models still inductively generalize these never-seen properties along their pre-learned taxonomic structure. What pre-training provides is therefore a reusable internal organization, rather than a set of individual memorized examples. However, when the prompt also implies a taxonomy that contradicts the one in the weights, the prompt wins (Figure 4.13).

6.2 Limitations

While our controlled study allows for a clean measurement, it also limits what it covers. Our taxonomy is small, balanced, and noiseless, while the real conceptual structure is large and complex to capture in full detail. Real categories overlap, admit exceptions, and depend on which property is being generalized. Nothing in our setup tests how induction behaves when the latent structure is ambiguous or when two structures compete within the same data, although Sections 4.5 and 4.6 make initial steps in this direction. Similarly, our properties are only verbs and adverbs stated in short templated phrases rather than in the varied and indirect ways properties might be expressed in natural text.

The models we tested range from 70M to 7B parameters, which is enough for an initial scaling comparison, but far from the scale of deployed systems in the real world. The trade-off we identify between memorization and generalization is one that should, in principle, grow more pronounced with size. But we cannot show that it continues, as other effects may dominate at larger scale.

6.3 Future directions

Establish the causal link. One extension would be to move from correlation to intervention. If the representational geometry is what drives induction, then editing it should change behavior in predictable ways. Ablating or steering along the directions that separate categories should selectively modify induction. Methods from the knowledge localization literature [18, 36, 63] give a starting point, and our setup is well suited to them because the ground-truth structure is known exactly.

Test for human-like regularities. Human induction is not only graded, but it also follows well-documented regularities. Examples include the *typicality* and *non-monotonicity* effects, as well as the *premise diversity*, where evidence from two distant entities supports a broader conclusion than evidence from two close ones [46]. Our framework can test each of these regularities directly,

since we control which entities appear as premises and where they are placed in the taxonomic tree. Finding that models reproduce these patterns would connect them to the cognitive literature more tightly than the broad claim that they generalize taxonomically.

Move toward realistic structure. Several relaxations of the synthetic setup are worth pursuing, such as unbalanced and deeper trees. These structures would test to what extent ground-truth taxonomies can be reflected in internal representations of models. Additionally, noisy or partially contradictory premises would test robustness to the inconsistency of real corpora. Overlapping taxonomies, where the correct grouping depends on the property being projected, would also test whether models can maintain more than one organization of the same entities at once. Finally, introducing pseudoword entities into an existing real category, such as a new bird, would test how new concepts can attach to structure the model already knows.

Track how in-context overridability changes. We showed that context can override the weights (Section 4.6), but not how that balance develops. If in-weights structure strengthens over training while the pull of context stays fixed, then overridability should decrease with more pre-training, and a heavily trained model would be harder to steer than a lightly trained one. This hypothesis is directly testable by repeating the conflict experiment across checkpoints, and it can provide insights on how much a prompt can be expected to redirect a large deployed model.

Extend the scaling study. A follow-up study could vary the data budget and model size together rather than one at a time as we do. If the trade-off is really about pressure to compress, then increasing the amount or diversity of taxonomic evidence should restore induction in larger models, and the point at which it does so might be practically useful to know.

6.4 Closing remarks

Taking all our experiments into account, we conclude that category-based property induction can develop naturally in LMs during pre-training. It is a consequence of training on data that has latent taxonomic structure, it is visible in the geometry of the model's representations, and it can be strengthened or weakened by choices in the training setup. For example, our results suggest longer pre-training strengthens the correct induction, while larger models at a fixed data budget do not. Additionally, a prior taxonomic structure inside a model may complicate replacement with a new taxonomy, and context can override the weights.

The result practitioners can take away is that strong recall and strong generalization are not necessarily the same thing, and that the conditions favoring one can work against the other. A model

that has memorized a domain perfectly may generalize poorly, and a standard evaluation could miss the difference. Understanding when a model will project what it has learned onto what it has not is a prerequisite for anticipating its behavior, and this thesis offers one controlled way to study it.

Bibliography

- [1] Zeyuan Allen-Zhu and Yuanzhi Li. “Physics of language models: part 3.1, knowledge storage and extraction”. In: *Proceedings of the 41st International Conference on Machine Learning*. ICML’24. Vienna, Austria: JMLR.org, 2024.
- [2] Usman Anwar, Abulhair Saparov, Javier Rando, Daniel Paleka, Miles Turpin, Peter Hase, Ekdeep Singh, Erik Jenner, Stephen Casper, Oliver Sourbut, Benjamin Edelman, Zhaowei Zhang, Mario Gunther, Anton Korinek, Jose Hernandez-Orallo, Lewis Hammond, Eric Bigelow, Alexander Pan, Lauro Langosco, Tomasz Korbak, Heidi Zhang, Ruiqi Zhong, Seán Ó hÉigearthaigh, Gabriel Rachet, Giulio Corsi, Alan Chan, Markus Anderljung, Lillian Edwards, Yoshua Bengio, Danqi Chen, Samuel Albanie, Tegan Maharaj, Jakob Foerster, Florian Tramer, He He, Atoosa Kasirzadeh, Yejin Choi, and David Krueger. “Foundational Challenges in Assuring Alignment and Safety of Large Language Models”. In: (2024).
- [3] Julia R. Badger and Laura R. Shapiro. “Evidence of a transition from perceptual to category induction in 3- to 9-year-old children”. In: *Journal of Experimental Child Psychology* 113.1 (2012), pp. 131–146. ISSN: 0022-0965. DOI: <https://doi.org/10.1016/j.jecp.2012.03.004>. URL: <https://www.sciencedirect.com/science/article/pii/S0022096512000562>.
- [4] Nora Belrose, Quintin Pope, Lucia Quirke, Alex Mallen, and Xiaoli Fern. “Neural networks learn statistics of increasing complexity”. In: *Proceedings of the 41st International Conference on Machine Learning*. ICML’24. Vienna, Austria: JMLR.org, 2024.
- [5] Lukas Berglund, Meg Tong, Maximilian Kaufmann, Mikita Balesni, Asa Cooper Stickland, Tomasz Korbak, and Owain Evans. “The Reversal Curse: LLMs trained on “A is B” fail to learn “B is A””. In: *The Twelfth International Conference on Learning Representations*. 2024. URL: <https://openreview.net/forum?id=GPKTIktA0k>.
- [6] Alexander Bick, Adam Blandin, and David J Deming. *The Rapid Adoption of Generative AI*. Working Paper 32966. National Bureau of Economic Research, Sept. 2024. DOI: 10.3386/w32966. URL: <http://www.nber.org/papers/w32966>.
- [7] Stella Biderman, Hailey Schoelkopf, Quentin Gregory Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, et al. “Pythia: A suite for analyzing large language models across training and scaling”. In: *International Conference on Machine Learning*. PMLR. 2023, pp. 2397–2430.

- [8] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel Ziegler, Jeffrey Wu, Clemens Winter, Chris Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. “Language Models are Few-Shot Learners”. In: *Advances in Neural Information Processing Systems*. Ed. by H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin. Vol. 33. Curran Associates, Inc., 2020, pp. 1877–1901. URL: https://proceedings.neurips.cc/paper_files/paper/2020/file/1457c0d6bfcb4967418bf8ac142f64a-Paper.pdf.
- [9] Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramer, and Chiyuan Zhang. “Quantifying Memorization Across Neural Language Models”. In: *The Eleventh International Conference on Learning Representations*. 2023. URL: https://openreview.net/forum?id=TatRHT_1cK.
- [10] Stephanie Chan, Adam Santoro, Andrew Lampinen, Jane Wang, Aaditya Singh, Pierre Richemond, James McClelland, and Felix Hill. “Data Distributional Properties Drive Emergent In-Context Learning in Transformers”. In: *Advances in Neural Information Processing Systems*. Ed. by S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh. Vol. 35. Curran Associates, Inc., 2022, pp. 18878–18891. DOI: 10.52202/068431-1371. URL: https://proceedings.neurips.cc/paper_files/paper/2022/file/77c6ccacfd9962e2307fc64680fc5ace-Conference.pdf.
- [11] Hoyeon Chang, Jinho Park, Seonghyeon Ye, Sohee Yang, Youngkyung Seo, Du-Seong Chang, and Minjoon Seo. “How Do Large Language Models Acquire Factual Knowledge During Pretraining?”. In: *The Thirty-eighth Annual Conference on Neural Information Processing Systems*. 2024. URL: <https://openreview.net/forum?id=TYdzj1EvBP>.
- [12] Aaron Chatterji, Thomas Cunningham, David J Deming, Zoe Hitzig, Christopher Ong, Carl Yan Shan, and Kevin Wadman. *How People Use ChatGPT*. Working Paper 34255. National Bureau of Economic Research, Sept. 2025. DOI: 10.3386/w34255. URL: <http://www.nber.org/papers/w34255>.
- [13] Cheng-Han Chiang, Sung-Feng Huang, and Hung-yi Lee. “Pretrained Language Model Embryology: The Birth of ALBERT”. In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Ed. by Bonnie Webber, Trevor Cohn, Yulan He, and Yang Liu. Online: Association for Computational Linguistics, Nov. 2020, pp. 6813–6828. DOI: 10.18653/v1/2020.emnlp-main.553. URL: <https://aclanthology.org/2020.emnlp-main.553/>.
- [14] Róbert Csordás, Kazuki Irie, and Juergen Schmidhuber. “The Devil is in the Detail: Simple Tricks Improve Systematic Generalization of Transformers”. In: *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. Ed. by Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih. Online and Punta Cana,

- Dominican Republic: Association for Computational Linguistics, Nov. 2021, pp. 619–634. DOI: 10.18653/v1/2021.emnlp-main.49. URL: <https://aclanthology.org/2021.emnlp-main.49/>.
- [15] Vitaly Feldman. “Does learning require memorization? a short tale about a long tail”. In: *Proceedings of the 52nd Annual ACM SIGACT Symposium on Theory of Computing*. STOC 2020. Chicago, IL, USA: Association for Computing Machinery, 2020, pp. 954–959. ISBN: 9781450369794. DOI: 10.1145/3357713.3384290. URL: <https://doi.org/10.1145/3357713.3384290>.
 - [16] Anna V Fisher and Vladimir M Sloutsky. “When Induction Meets Memory: Evidence for Gradual Transition From Similarity-Based to Category-Based Induction”. In: *Child Development* 76.3 (May 2005), pp. 583–597. ISSN: 0009-3920. DOI: 10.1111/j.1467-8624.2005.00865.x. eprint: <https://academic.oup.com/chidev/article-pdf/76/3/583/64506517/j.1467-8624.2005.00865.x.pdf>. URL: <https://doi.org/10.1111/j.1467-8624.2005.00865.x>.
 - [17] Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, Shawn Presser, and Connor Leahy. “The Pile: An 800GB Dataset of Diverse Text for Language Modeling”. In: *arXiv preprint arXiv:2101.00027* (2020).
 - [18] Mor Geva, Jasmijn Bastings, Katja Filippova, and Amir Globerson. “Dissecting Recall of Factual Associations in Auto-Regressive Language Models”. In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Ed. by Houda Bouamor, Juan Pino, and Kalika Bali. Singapore: Association for Computational Linguistics, Dec. 2023, pp. 12216–12235. DOI: 10.18653/v1/2023.emnlp-main.751. URL: <https://aclanthology.org/2023.emnlp-main.751/>.
 - [19] Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. “Transformer feed-forward layers are key-value memories”. In: *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. 2021, pp. 5484–5495.
 - [20] MohammadTaghi Hajiaghayi and Marina Knittel. “Improved hierarchical clustering on massive datasets with broad guarantees”. In: *arXiv preprint arXiv:2101.04818* (2021).
 - [21] Simon Jerome Han, Keith J. Ransom, Andrew Perfors, and Charles Kemp. “Inductive reasoning in humans and large language models”. In: *Cognitive Systems Research* 83 (2024), p. 101155. ISSN: 1389-0417. DOI: <https://doi.org/10.1016/j.cogsys.2023.101155>. URL: <https://www.sciencedirect.com/science/article/pii/S1389041723000839>.
 - [22] Evan Heit. “Properties of inductive reasoning”. In: *Psychonomic Bulletin & Review* 7.4 (Dec. 2000), pp. 569–592. ISSN: 1531-5320. DOI: 10.3758/BF03212996. URL: <https://doi.org/10.3758/BF03212996>.

- [23] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, et al. “Training compute-optimal large language models”. In: *arXiv preprint arXiv:2203.15556* (2022).
- [24] Jesse Hoogland, George Wang, Matthew Farrugia-Roberts, Liam Carroll, Susan Wei, and Daniel Murfet. “Loss Landscape Degeneracy and Stagewise Development in Transformers”. In: *Transactions on Machine Learning Research* (2025). ISSN: 2835-8856. URL: <https://openreview.net/forum?id=45qJyBG80j>.
- [25] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. “Scaling laws for neural language models”. In: *arXiv preprint arXiv:2001.08361* (2020).
- [26] F.C. Keil. *Semantic and Conceptual Development: An Ontological Perspective*. Cognitive science series. Harvard University Press, 1979. ISBN: 9780674801004. URL: <https://books.google.fr/books?id=k2V-AAAAMAAJ>.
- [27] Charles Kemp and Joshua B Tenenbaum. “Structured statistical models of inductive reasoning”. en. In: *Psychol Rev* 116.1 (Jan. 2009), pp. 20–58.
- [28] Charles Kemp and Joshua B. Tenenbaum. “The discovery of structural form”. In: *Proceedings of the National Academy of Sciences* 105.31 (2008), pp. 10687–10692. DOI: 10.1073/pnas.0802631105. eprint: <https://www.pnas.org/doi/pdf/10.1073/pnas.0802631105>. URL: <https://www.pnas.org/doi/abs/10.1073/pnas.0802631105>.
- [29] Emmanuel Keuleers and Marc Brysbaert. “Wuggy: A multilingual pseudoword generator”. In: *Behavior Research Methods* 42.3 (Aug. 2010), pp. 627–633. ISSN: 1554-3528. DOI: 10.3758/BRM.42.3.627. URL: <https://doi.org/10.3758/BRM.42.3.627>.
- [30] Jiyeon Kim, Hyunji Lee, Hyowon Cho, Joel Jang, Hyeonbin Hwang, Seungpil Won, Youbin Ahn, Dohaeng Lee, and Minjoon Seo. “Knowledge Entropy Decay during Language Model Pretraining Hinders New Knowledge Acquisition”. In: *The Thirteenth International Conference on Learning Representations*. 2025. URL: <https://openreview.net/forum?id=eHehzSDUFp>.
- [31] Diederik P. Kingma and Jimmy Ba. “Adam: A Method for Stochastic Optimization”. In: *International Conference on Learning Representations*. 2015.
- [32] Andrew Kyle Lampinen, Arslan Chaudhry, Stephanie C.Y. Chan, Cody Wild, Diane Wan, Alexander Ku, Jörg Bornschein, Razvan Pascanu, Murray Shanahan, and James Lloyd McClelland. “On the generalization of language models from in-context learning and finetuning: a controlled study”. In: *First Workshop on Foundations of Reasoning in Language Models*. 2025. URL: <https://openreview.net/forum?id=nbENxsEmxn>.

- [33] Leo Z. Liu, Yizhong Wang, Jungo Kasai, Hannaneh Hajishirzi, and Noah A. Smith. “Probing Across Time: What Does RoBERTa Know and When?” In: *Findings of the Association for Computational Linguistics: EMNLP 2021*. Ed. by Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih. Punta Cana, Dominican Republic: Association for Computational Linguistics, Nov. 2021, pp. 820–842. DOI: 10.18653/v1/2021.findings-emnlp.71. URL: <https://aclanthology.org/2021.findings-emnlp.71/>.
- [34] J M Mandler. “How to build a baby: II. Conceptual primitives”. en. In: *Psychological Review* 99.4 (Oct. 1992), pp. 587–604.
- [35] Jean M. Mandler and Laraine McDonough. “Concept formation in infancy”. In: *Cognitive Development* 8.3 (1993), pp. 291–318. ISSN: 0885-2014. DOI: [https://doi.org/10.1016/S0885-2014\(93\)80003-C](https://doi.org/10.1016/S0885-2014(93)80003-C). URL: <https://www.sciencedirect.com/science/article/pii/S088520149380003C>.
- [36] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. “Locating and Editing Factual Associations in GPT”. In: *Advances in Neural Information Processing Systems*. Ed. by S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh. Vol. 35. Curran Associates, Inc., 2022, pp. 17359–17372. DOI: 10.52202/068431-1262. URL: https://proceedings.neurips.cc/paper_files/paper/2022/file/6f1d43d5a82a37e89b0665b33bf3a182-Paper-Conference.pdf.
- [37] Sewon Min, Xinxi Lyu, Ari Holtzman, Mikel Artetxe, Mike Lewis, Hannaneh Hajishirzi, and Luke Zettlemoyer. “Rethinking the Role of Demonstrations: What Makes In-Context Learning Work?” In: *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. Ed. by Yoav Goldberg, Zornitsa Kozareva, and Yue Zhang. Abu Dhabi, United Arab Emirates: Association for Computational Linguistics, Dec. 2022, pp. 11048–11064. DOI: 10.18653/v1/2022.emnlp-main.759. URL: <https://aclanthology.org/2022.emnlp-main.759/>.
- [38] Kanishka Misra, Julia Rayz, and Allyson Ettinger. “A Property Induction Framework for Neural Language Models”. In: *Proceedings of the 44th Annual Conference of the Cognitive Science Society*. Ed. by J. Culbertson, A. Perfors, H. Rabagliati, and V. Ramenzoni. Cognitive Science Society, 2022. URL: <https://escholarship.org/uc/item/6170h6nj>.
- [39] Kanishka Misra, Julia Rayz, and Allyson Ettinger. “COMPS: Conceptual Minimal Pair Sentences for testing Robust Property Knowledge and its Inheritance in Pre-trained Language Models”. In: *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*. Ed. by Andreas Vlachos and Isabelle Augenstein. Dubrovnik, Croatia: Association for Computational Linguistics, May 2023, pp. 2928–2949. DOI: 10.18653/v1/2023.eacl-main.213. URL: <https://aclanthology.org/2023.eacl-main.213/>.
- [40] Max Müller-Eberstein, Rob van der Goot, Barbara Plank, and Ivan Titov. “Subspace Chronicles: How Linguistic Information Emerges, Shifts and Interacts during Language Model Training”. In: *Findings of the Association for Computational Linguistics: EMNLP 2023*. Ed. by Houda Bouamor, Juan Pino, and Kalika Bali. Singapore: Association for Computational Linguistics,

- Dec. 2023, pp. 13190–13208. DOI: 10.18653/v1/2023.findings-emnlp.879. URL: <https://aclanthology.org/2023.findings-emnlp.879/>.
- [41] Gregory L. Murphy. *The Big Book of Concepts*. The MIT Press, July 2002. ISBN: 9780262280358. DOI: 10.7551/mitpress/1602.001.0001. URL: <https://doi.org/10.7551/mitpress/1602.001.0001>.
 - [42] Andres Nava and Matthieu Wyart. *Hierarchical Concept Geometry in Language Models Emerges from Word Co-occurrence*. 2026. arXiv: 2605.23821 [cs.CL]. URL: <https://arxiv.org/abs/2605.23821>.
 - [43] Nostalgebraist. “Interpreting GPT: The logit lens”. In: *Blog Post* (2020). URL: <https://www.lesswrong.com/posts/AcKRB8wDpdAN6v6ru/interpreting-gpt-the-logit-lens>.
 - [44] Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, Nathan Lambert, Dustin Schwenk, Oyvind Tafjord, Taira Anderson, David Atkinson, Faeze Brahman, Christopher Clark, Pradeep Dasigi, Nouha Dziri, Michal Guerquin, Hamish Ivison, Pang Wei Koh, Jiacheng Liu, Saumya Malik, William Merrill, Lester James V. Miranda, Jacob Morrison, Tyler Murray, Crystal Nam, Valentina Pyatkin, Aman Rangapur, Michael Schmitz, Sam Skjonsberg, David Wadden, Christopher Wilhelm, Michael Wilson, Luke Zettlemoyer, Ali Farhadi, Noah A. Smith, and Hannaneh Hajishirzi. *2 OLMo 2 Furious*. 2024. arXiv: 2501.00656 [cs.CL]. URL: <https://arxiv.org/abs/2501.00656>.
 - [45] Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Scott Johnston, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. *In-context Learning and Induction Heads*. 2022. arXiv: 2209.11895 [cs.LG]. URL: <https://arxiv.org/abs/2209.11895>.
 - [46] Daniel Osherson, Edward Smith, Ormond Wilkie, Alejandro López-Rousseau, and Eldar Shafir. “Category-Based Induction”. In: *Psychological Review* 97 (Apr. 1990), pp. 185–200. DOI: 10.1037/0033-295X.97.2.185.
 - [47] Kiho Park, Yo Joong Choe, Yibo Jiang, and Victor Veitch. “The Geometry of Categorical and Hierarchical Concepts in Large Language Models”. In: *International Conference on Learning Representations*. Ed. by Y. Yue, A. Garg, N. Peng, F. Sha, and R. Yu. Vol. 2025. 2025, pp. 76441–76463. URL: https://proceedings.iclr.cc/paper_files/paper/2025/file/be7430d22a4dae8516894e32f2fcc6db-Paper-Conference.pdf.
 - [48] Kiho Park, Yo Joong Choe, and Victor Veitch. “The Linear Representation Hypothesis and the Geometry of Large Language Models”. In: *Causal Representation Learning Workshop at NeurIPS 2023*. 2023. URL: <https://openreview.net/forum?id=T0Po0Jg8cK>.
 - [49] Sabina Pauen. “Evidence for knowledge-based category discrimination in infancy”. en. In: *Child Dev* 73.4 (July 2002), pp. 1016–1033.

- [50] Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. “Grokking: Generalization beyond overfitting on small algorithmic datasets”. In: *arXiv preprint arXiv:2201.02177* (2022).
- [51] Alec Radford, Jeff Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. “Language Models are Unsupervised Multitask Learners”. In: (2019).
- [52] Maria Refinetti, Alessandro Ingrosso, and Sebastian Goldt. “Neural networks trained with SGD learn distributions of increasing complexity”. In: *Proceedings of the 40th International Conference on Machine Learning*. Ed. by Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett. Vol. 202. Proceedings of Machine Learning Research. PMLR, July 2023, pp. 28843–28863. URL: <https://proceedings.mlr.press/v202/refinetti23a.html>.
- [53] Lance J. Rips. “Inductive judgments about natural categories”. In: *Journal of Verbal Learning and Verbal Behavior* 14.6 (1975), pp. 665–681. ISSN: 0022-5371. DOI: [https://doi.org/10.1016/S0022-5371\(75\)80055-7](https://doi.org/10.1016/S0022-5371(75)80055-7). URL: <https://www.sciencedirect.com/science/article/pii/S0022537175800557>.
- [54] Juan Diego Rodriguez, Aaron Mueller, and Kanishka Misra. “Characterizing the Role of Similarity in the Property Inferences of Language Models”. In: *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. Ed. by Luis Chiruzzo, Alan Ritter, and Lu Wang. Albuquerque, New Mexico: Association for Computational Linguistics, Apr. 2025, pp. 11515–11533. ISBN: 979-8-89176-189-6. DOI: 10.18653/v1/2025.naacl-long.574. URL: <https://aclanthology.org/2025.naacl-long.574/>.
- [55] Timothy T. Rogers and James L. McClelland. *Semantic Cognition: A Parallel Distributed Processing Approach*. The MIT Press, June 2004. ISBN: 9780262282505. DOI: 10.7551/mitpress/6161.001.0001. URL: <https://doi.org/10.7551/mitpress/6161.001.0001>.
- [56] Naomi Saphra and Adam Lopez. “Understanding Learning Dynamics Of Language Models with SVCCA”. In: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. Ed. by Jill Burstein, Christy Doran, and Thamar Solorio. Minneapolis, Minnesota: Association for Computational Linguistics, June 2019, pp. 3257–3267. DOI: 10.18653/v1/N19-1329. URL: <https://aclanthology.org/N19-1329/>.
- [57] Andrew M. Saxe, James L. McClelland, and Surya Ganguli. “A mathematical theory of semantic development in deep neural networks”. In: *Proceedings of the National Academy of Sciences* 116.23 (2019), pp. 11537–11546. DOI: 10.1073/pnas.1820226116. eprint: <https://www.pnas.org/doi/pdf/10.1073/pnas.1820226116>. URL: <https://www.pnas.org/doi/abs/10.1073/pnas.1820226116>.

- [58] Rico Sennrich, Barry Haddow, and Alexandra Birch. “Neural Machine Translation of Rare Words with Subword Units”. In: *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Ed. by Katrin Erk and Noah A. Smith. Berlin, Germany: Association for Computational Linguistics, Aug. 2016, pp. 1715–1725. DOI: 10.18653/v1/P16-1162. URL: <https://aclanthology.org/P16-1162/>.
- [59] Oscar Skenan, Md Rifat Arefin, Dan Zhao, Niket Patel, Jalal Naghiyev, Yann LeCun, and Ravid Shwartz-Ziv. “Layer by layer: Uncovering hidden representations in language models”. In: *arXiv preprint arXiv:2502.02013* (2025).
- [60] Steven Sloman. “Feature-Based Induction”. In: *Cognitive Psychology* 25 (Aug. 1993). DOI: 10.1006/cogp.1993.1006.
- [61] Kushal Tirumala, Aram H. Markosyan, Luke Zettlemoyer, and Armen Aghajanyan. “Memorization Without Overfitting: Analyzing the Training Dynamics of Large Language Models”. In: *Advances in Neural Information Processing Systems*. Ed. by Alice H. Oh, Alekh Agarwal, Danielle Belgrave, and Kyunghyun Cho. 2022. URL: <https://openreview.net/forum?id=u3vEuRr08MT>.
- [62] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. “Attention is All you Need”. In: *Advances in Neural Information Processing Systems*. Ed. by I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett. Vol. 30. Curran Associates, Inc., 2017. URL: https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf.
- [63] Song Wang, Yaochen Zhu, Haochen Liu, Zaiyi Zheng, Chen Chen, and Jundong Li. “Knowledge Editing for Large Language Models: A Survey”. In: *ACM Comput. Surv.* 57.3 (Nov. 2024). ISSN: 0360-0300. DOI: 10.1145/3698590. URL: <https://doi.org/10.1145/3698590>.
- [64] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. “Chain-of-thought prompting elicits reasoning in large language models”. In: *Proceedings of the 36th International Conference on Neural Information Processing Systems*. NIPS ’22. New Orleans, LA, USA: Curran Associates Inc., 2022. ISBN: 9781713871088.
- [65] Sang Michael Xie, Aditi Raghunathan, Percy Liang, and Tengyu Ma. “An Explanation of In-context Learning as Implicit Bayesian Inference”. In: *International Conference on Learning Representations*. 2022. URL: <https://openreview.net/forum?id=RdJVFCHjUMI>.
- [66] Chiyuan Zhang, Samy Bengio, Moritz Hardt, Benjamin Recht, and Oriol Vinyals. “Understanding deep learning requires rethinking generalization”. In: *International Conference on Learning Representations*. 2017. URL: <https://openreview.net/forum?id=Sy8gdB9xx>.

- [67] Nicolas Zucchet, Jorg Bornschein, Stephanie C.Y. Chan, Andrew Kyle Lampinen, Razvan Pascanu, and Soham De. “How do language models learn facts? Dynamics, curricula and hallucinations”. In: *Second Conference on Language Modeling*. 2025. URL: <https://openreview.net/forum?id=vBcGnragkr>.